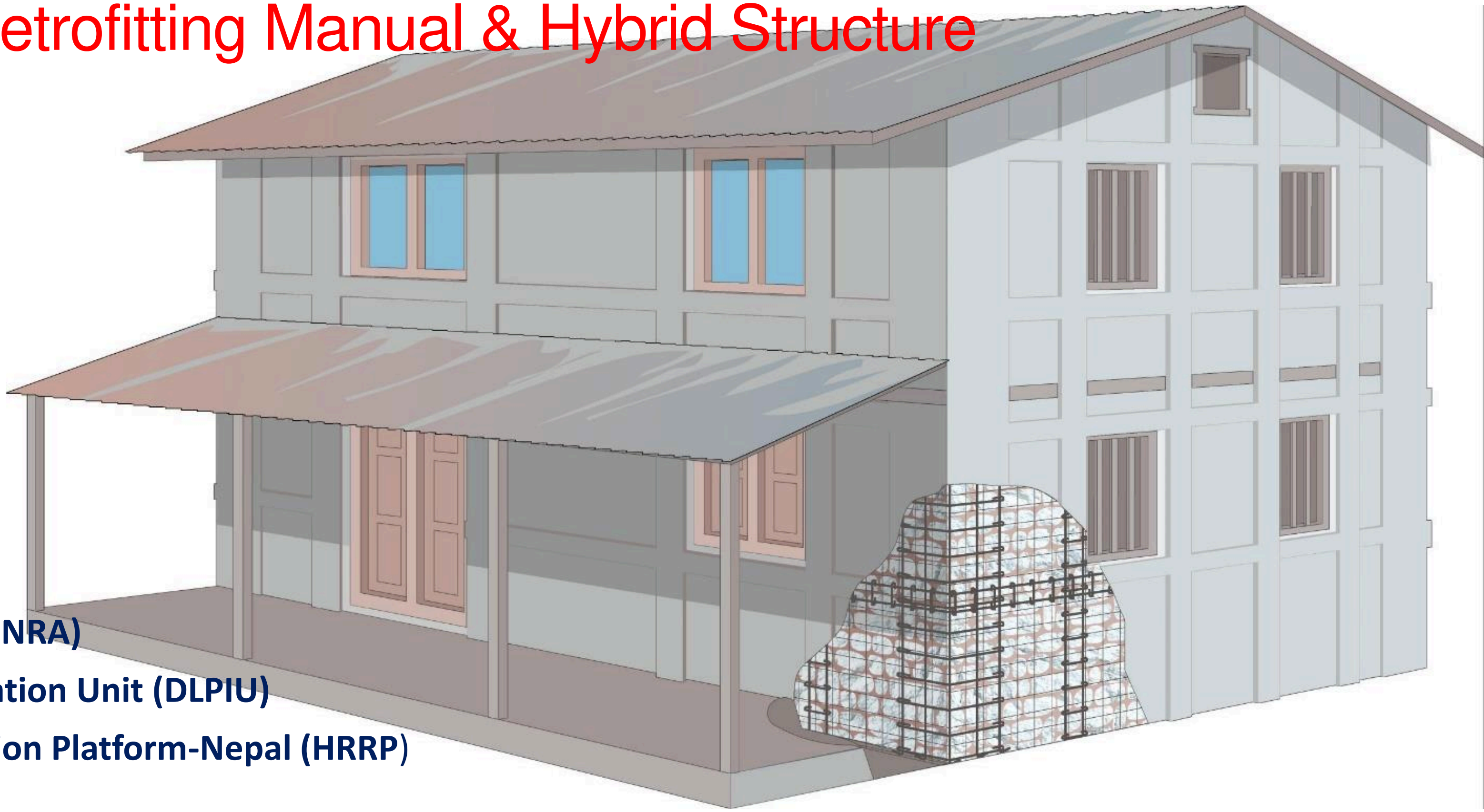
Housing Recovery and Reconstruction Platform-Nepal (HRRP)

Orientation Program

on

Repair & Retrofitting Manual & Hybrid Structure



Organized by:

National Reconstruction Authority(NRA)

MOUD/District Project Implementation Unit (DLPIU)

Housing Recovery and Reconstruction Platform-Nepal (HRRP)

Background

- Almost 25,000 households across the 31 earthquake affected district were categorized as damage grade 2 of major technical solution or damage grade 3 of minor technical solution

- Under the Government of Nepal (GoN) housing reconstruction program, there is provision of housing retrofit grant of 100,000 NPRs if its complies with the relevant standards and specifications

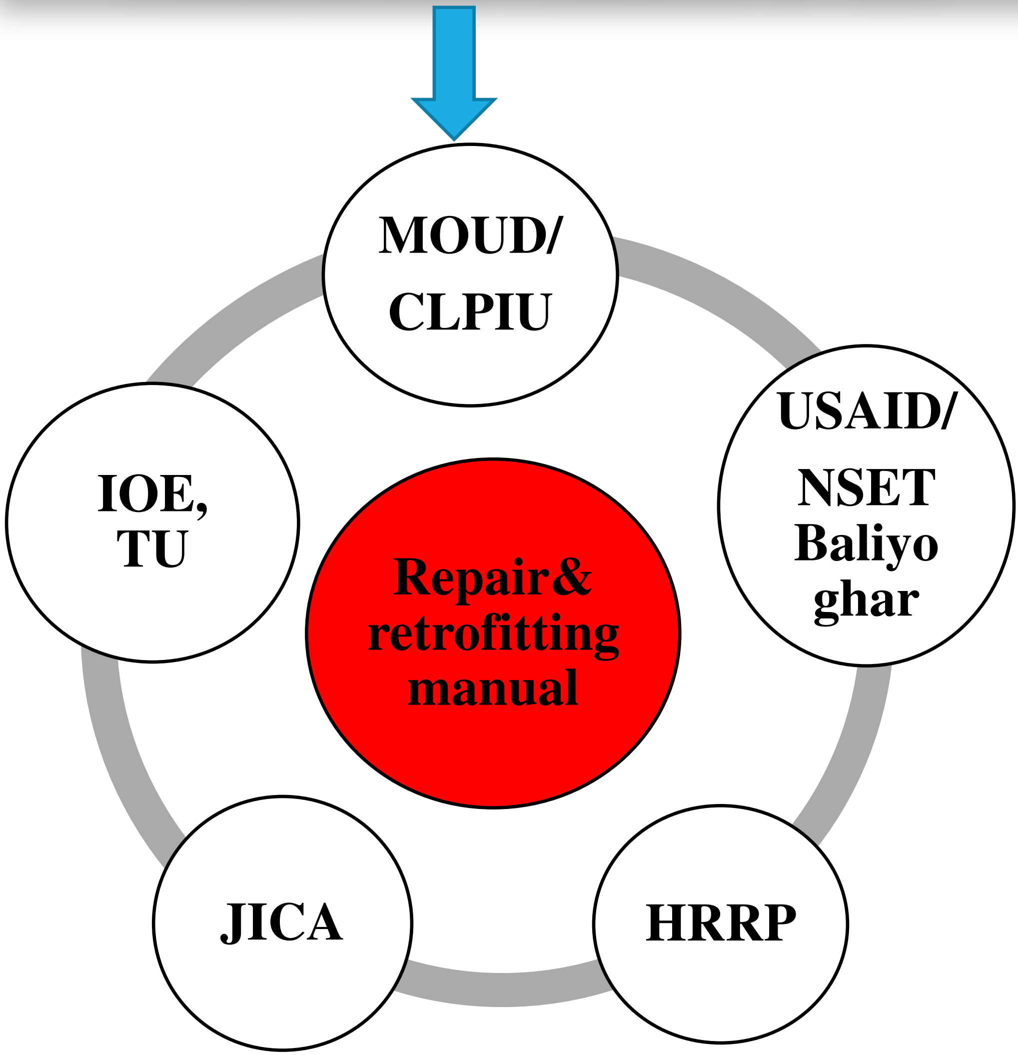
- This manual outlines these standards and specifications as well as the minimum intervention works required to carry out the retrofitting

Objective of Manual

- To set the minimum criteria to provide the tranches under the retrofitting grant
- Cover policies for distribution of tranches with minimum technical intervention options
- To support engineers for inspection, help them to provide advice and guidance to households

Initiatives

Standardization Committee of NRA



Scope of Manual

Applicability

- The repair and retrofitting strategies set forth in this manual are applicable only for residential houses categorized as damage grade 2 (major) or 3 (minor) after Gorkha earthquake 2015.

- The designs mentioned in the manual are ready-to-use designs for all structural components, but some provisions mentioned are set as advisory measures.

Classification of Building, Building Act 2055

- **Category "A"**: Modern building to be built, based on the international state-of-the-art, also in pursuance of the building codes to be followed in developed countries.
- **Category "B"**:Buildings with plinth area of more than one Thousand square feet, with more than Three floors including the ground floor or with structural span of more than 4.5 meters.
- **Category "C"** :Buildings with plinth area of up to One Thousand square feet, with up to Three floors including the ground floor or with structural span of up to 4.5 meters.
- **Category "D"** : Small Houses, sheds made of baked or unbaked brick, Stone ,Clay, bamboos, grass etc., except those set forth clause (a), (b) & (c)

Limitation

- The repair and retrofitting strategies are only for damaged non-engineered residential buildings falls under fall under category ‘C’ and ‘D’ of NBC.

If the intervention has already been completed as per, or similar to, the strategies outlined in this manual, Government of Nepal published documents, or as per international practices, and are based on codal provision ensuring life safety with quality construction, then applications can be forwarded only after thorough engineering judgement

Grant Process

GoN

Beneficiary

Beneficiary Agreement for Repair and Retrofit

First tranche (NRs. 50k)

Exiting Building Inspection and Recommendation for Intervention works

Final Inspection : After Repair and Retrofitting completes (Before Plaster & Finishing)

Second tranche : NRs. 50k

General Terminology

Repair:

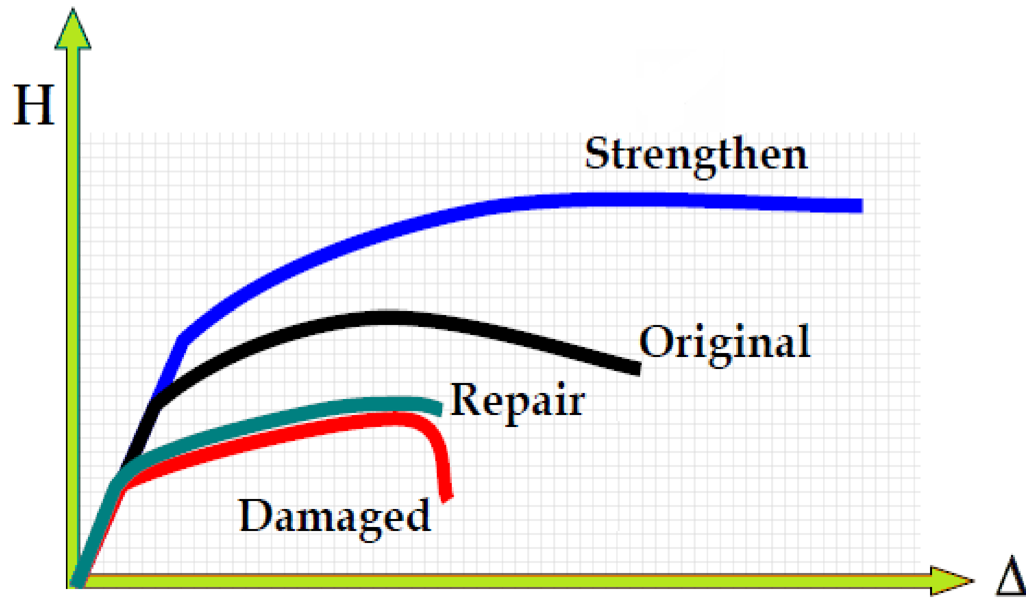
To bring back the architectural shape of the building so that all services start working & the functioning of building is resumed quickly.

Restoration:

To carry out structural repairs to load bearing components to restore its original strength.

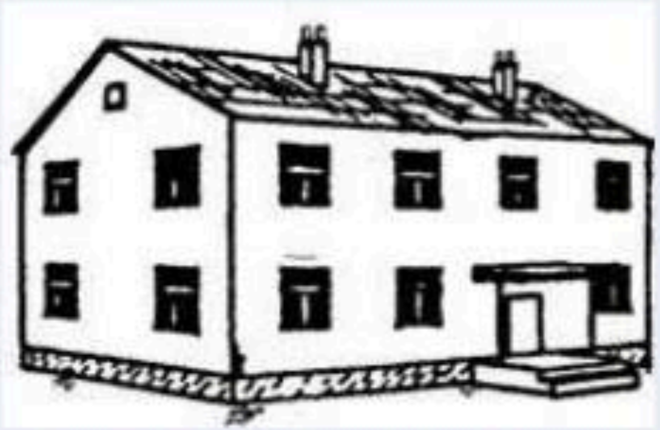
Retrofitting:

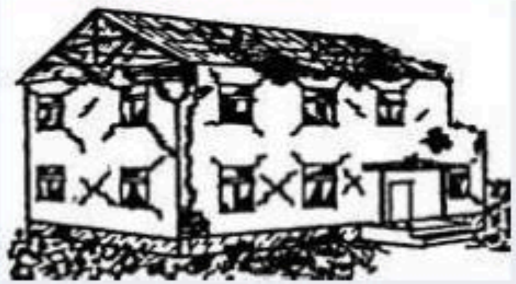
To make buildings stronger than before.



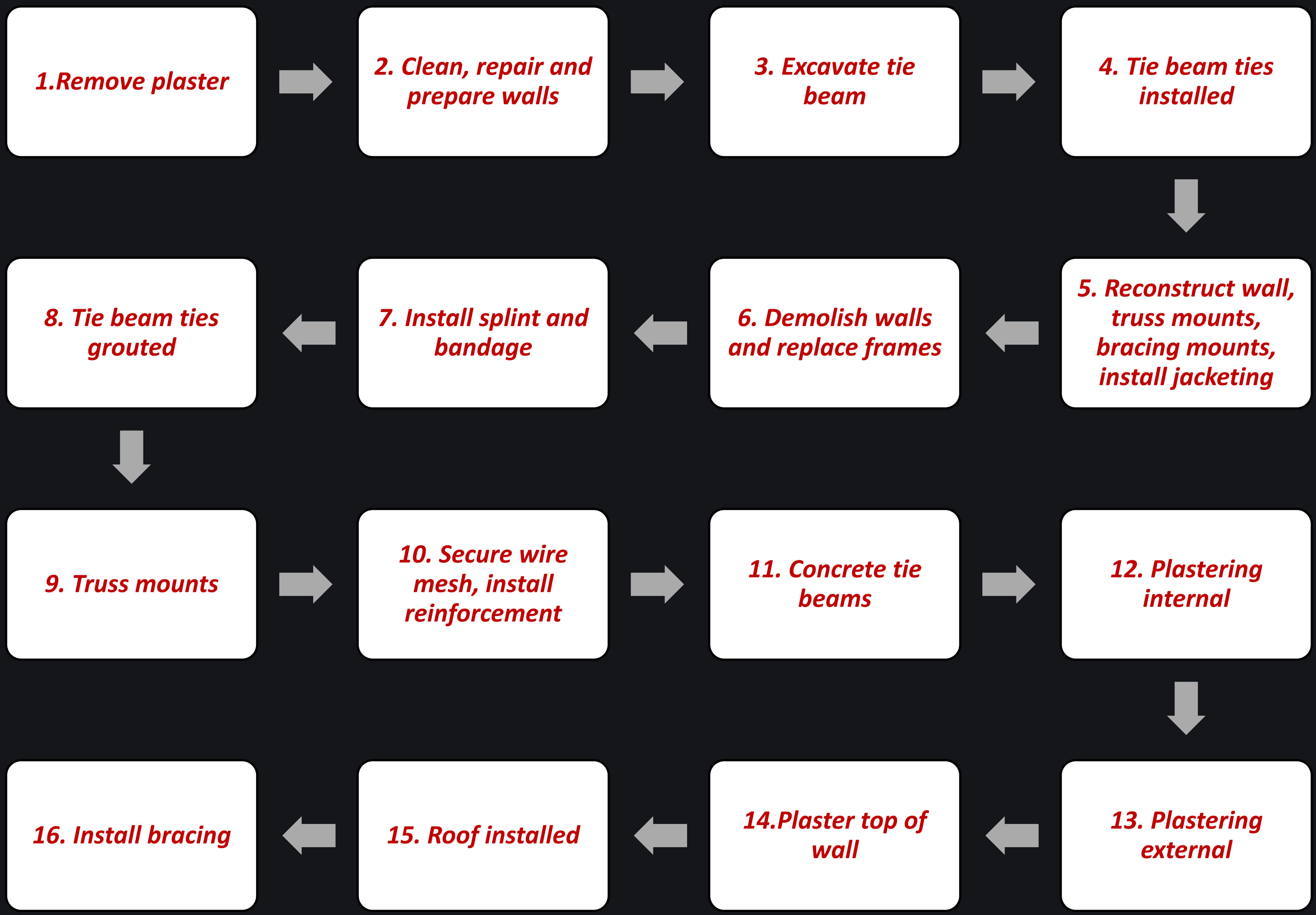
MASONRY STRUCTURE

Damage Grade: Masonry Buildings

Classification of damage to masonry buildings	
 Grade 1: Negligible to slight damage	Structural damage : No Non-structural damage: Slight • Hair-line cracks in very few walls. • Fall of small pieces of plaster only. • Fall of loose stones from upper parts of buildings in very few cases.
 Grade 2: Moderate damage	Structural damage : Slight Non-structural damage: Moderate • Cracks in many walls. • Fall of fairly large pieces of plaster. • Partial collapse of chimneys.

 Grade 3: Substantial to heavy damage	Structural damage: Moderate Non-structural damage: Heavy • Large and extensive cracks in most walls. • Roof tiles detach. • Chimneys fracture at the roof line; failure of individual non-structural elements (partitions, gable walls).
 Grade 4: Very heavy damage	Structural damage: Heavy Non-structural damage: Very heavy • Serious failure of walls; partial structural failure of roofs and floors.
 Grade 5: Destruction	Structural damage: very heavy • Total or near total collapse.

Retrofitting process



Contents

PART-A: Seismic Damage and Intervention

PART-A: Seismic Damage and Intervention

This part deals with seismic damages and possible intervention that needs to turn the building in to pre-earthquake condition.

Its details in Part D Construction sequences

1. Foundation damage & mitigation work

   [Key problem, repair solution and retrofit solution]

2. Roof/Floor partial collapse and Mitigation Work

   [Key problem, repair solution and retrofit solution]

3. Cracks in wall and Mitigation Work

   [Key problem, repair solution and retrofit solution]

1.Foundation damage & mitigation work

Potential damages in foundation are as follows :

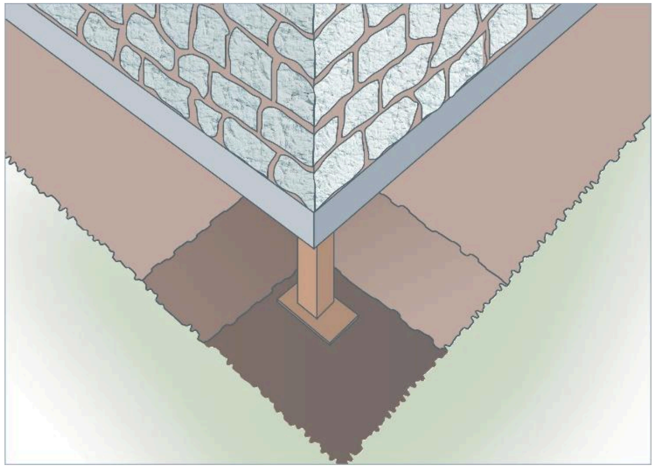
- F.1 Cracked stone masonry
- F.2 Settlement of Foundation
- F.3 Bulged Stone Walls
- F.4 Dislocations and Loose Stone
- F.5 Stone foundation wall interruption, removal of portions of the wall, & loss of structural integrity

Foundation damage & mitigation work Contd.

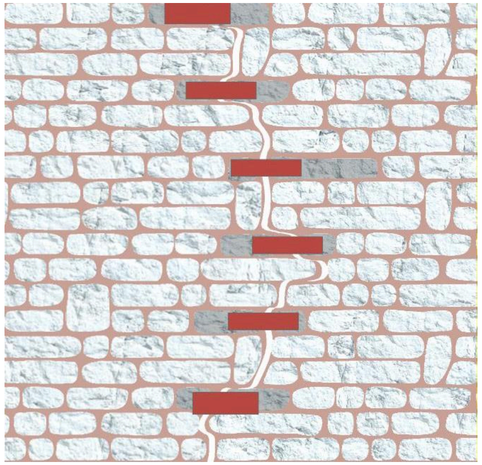
- 

F.3 Bulged and leaning stone Foundation

F.2Settlement of Foundation



Reconstruction of Foundation with proper safety



Repair Foundation

Roof/Floor partial collapse and Mitigation Work

Potential damages in floor and/or roof are as follows :

- R.1: Partial to heavy damage on gable wall
- R.2 : Sliding of roof materials (stone slate or clay tiles)
- R.3 : Roof connections failure
- R.4 : Floor connection
- R.5 : Floor to wall connection
- R.6 : Roof to wall connection
- R.7 :Wall to wall connection

R1.Partial to heavy damage of Gable wall

R1.1 Gable wall toppling

R.1.2 Gable wall toppling

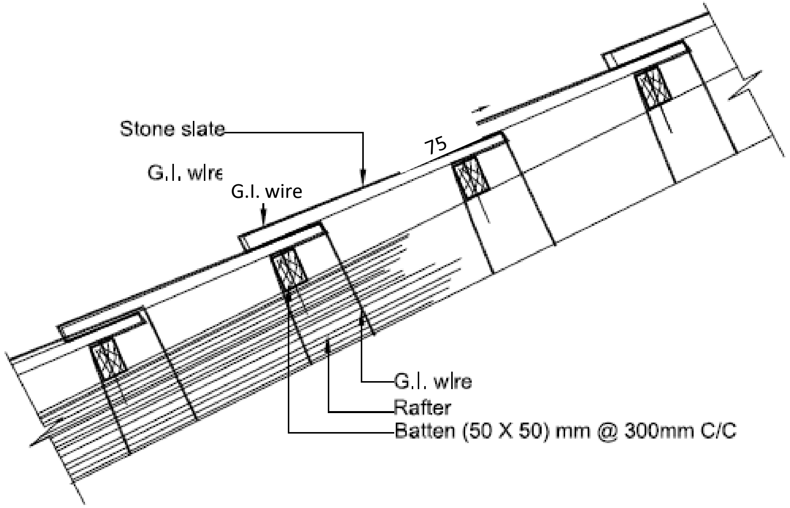
Solution : Remove heavy material replace with the lighter materials

Use of C.G.I. sheet at Gable part

Use of timber planks at Gable part

R.2:Sliding of Roofing materials

Solution



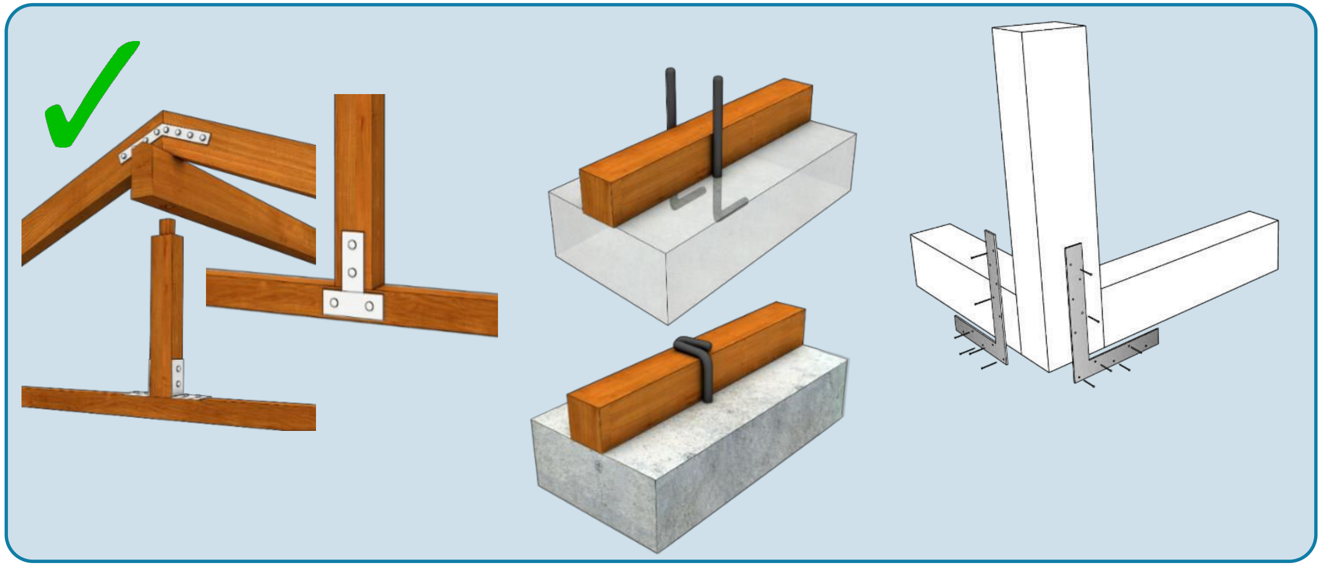
Detail of anchoring slate stone
Source: NBC203 2015, P44

Detail of fixing clay tile
Source: NBC203 2015, P45

R.3:Roof connection failure

R.3.1 Purlin detached from rafter due to inadequate nailing

R.3.2 Connection of wooden truss



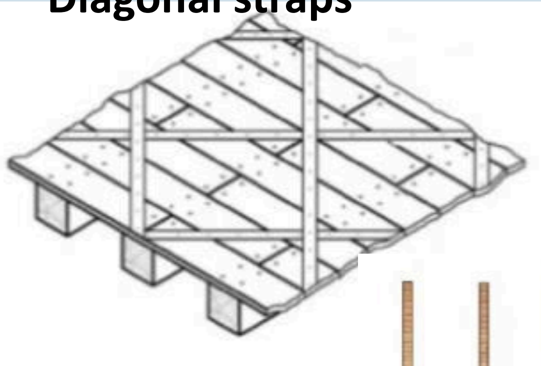
R.4: Floor connection

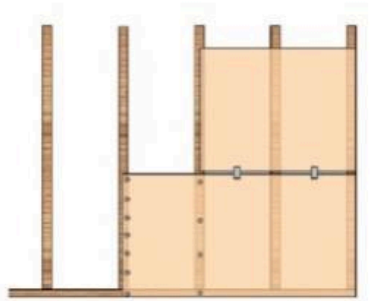
rafters R.4.1 view inside the attic showing the and the absence of collar beams and joist connections to the rafters

R.4.2 Inadequate and poor floor members

Solution

Diagonal straps

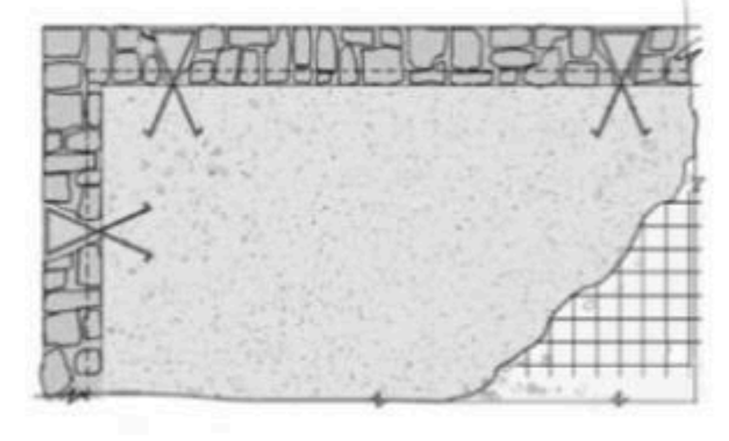




Nailed sheeting

Connectivity of roof member

New RC Floor



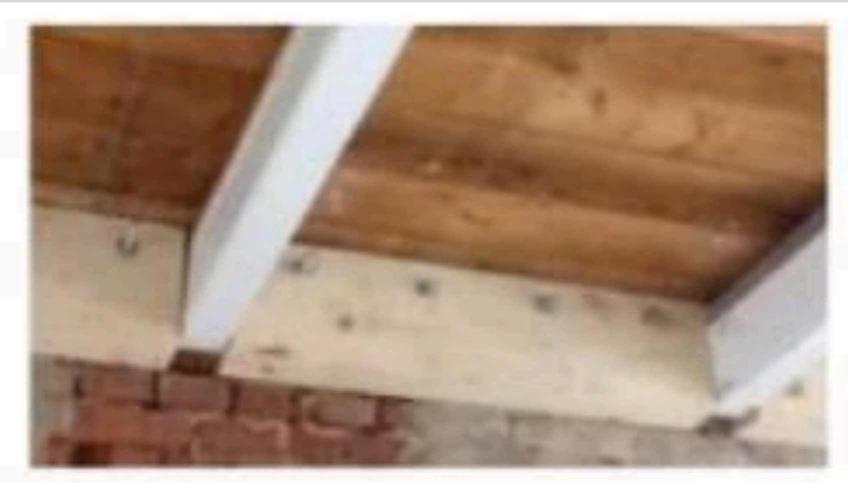
R.5: Floor to wall connection

Solution

- R.5.1 Floor joists supported by half of the wall width (insufficient anchorage)

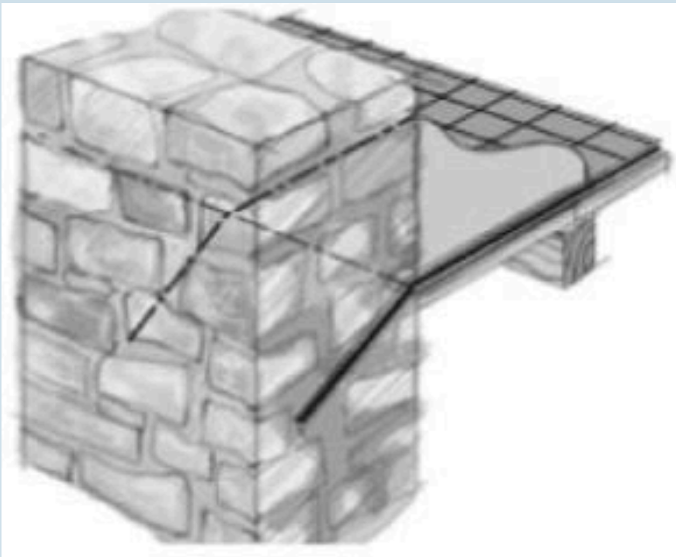
R.5.2Wall-to-floor connection parallel to the joists (insufficient anchorage)

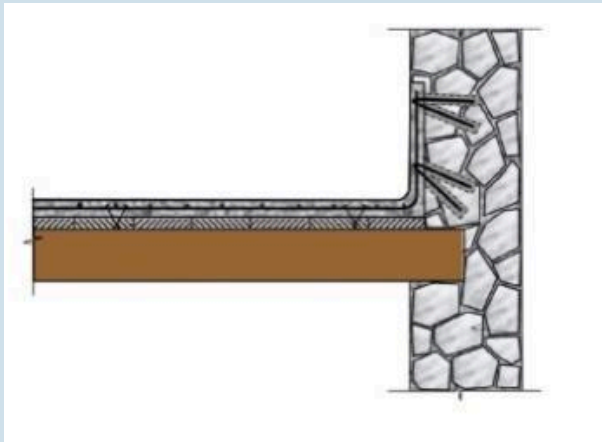
Anchored with splint and bandage at floor level



R.6:Roof to wall connection

Solution



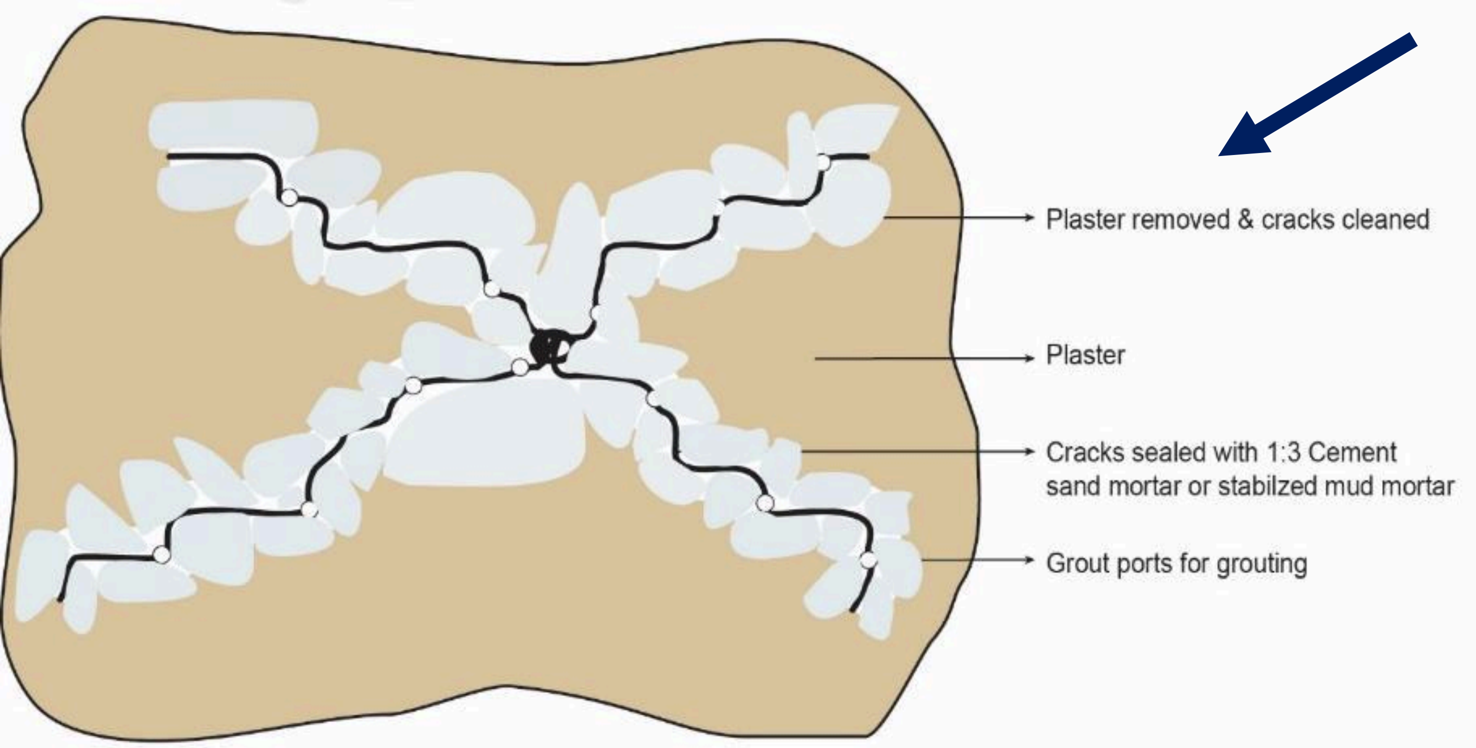


3. Cracks in wall and mitigation works

Potential damages in structural and non-structural wall are as follows :

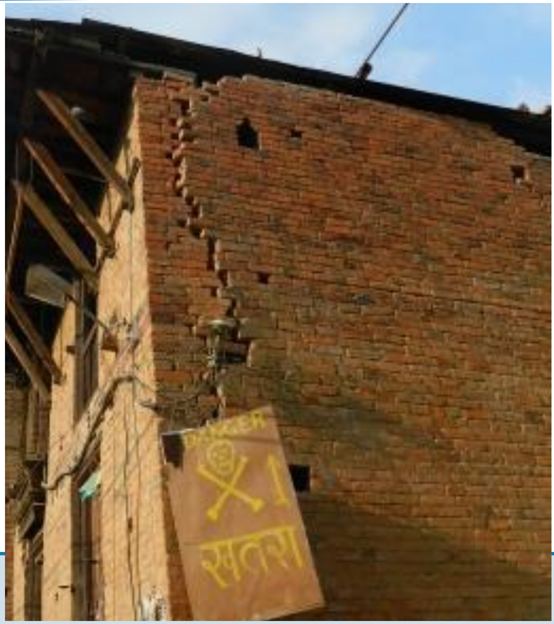
- C.1 : Minor cracks
- C.2 : Major cracks
- C.3 : Heavy cracks
- C.4 : Out-of-plane failure of walls
- C.5 : Wall to wall connection

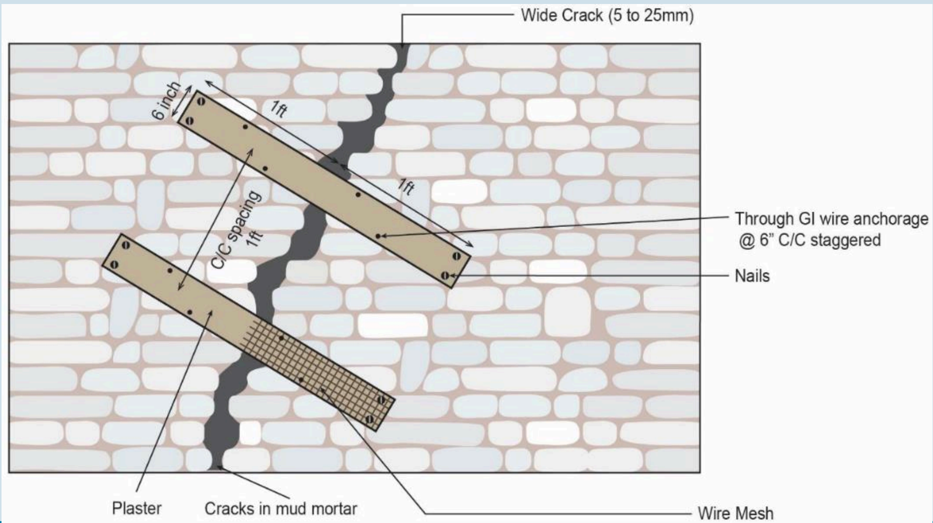
C.1:Minor Cracks

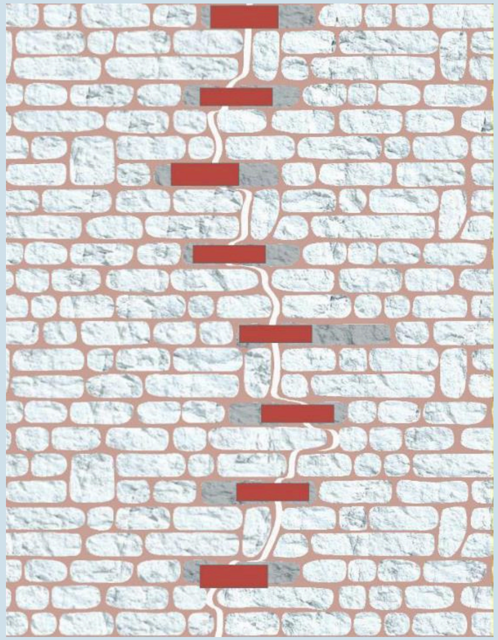


Repair Minor To Cracks Using Grouting

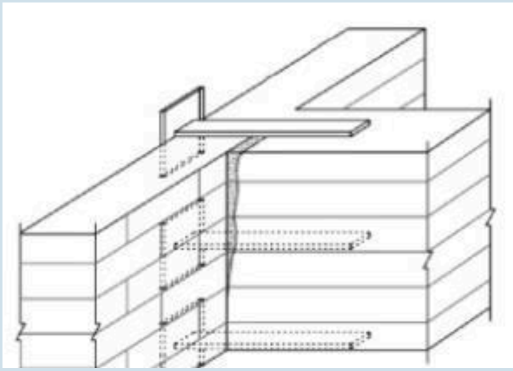
Major to heavy Crack



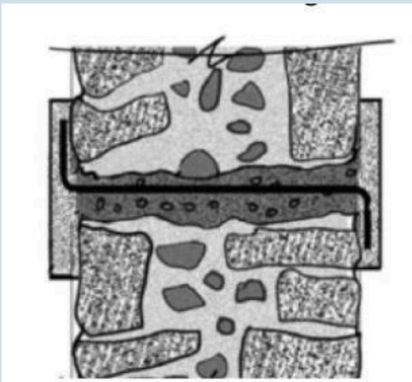




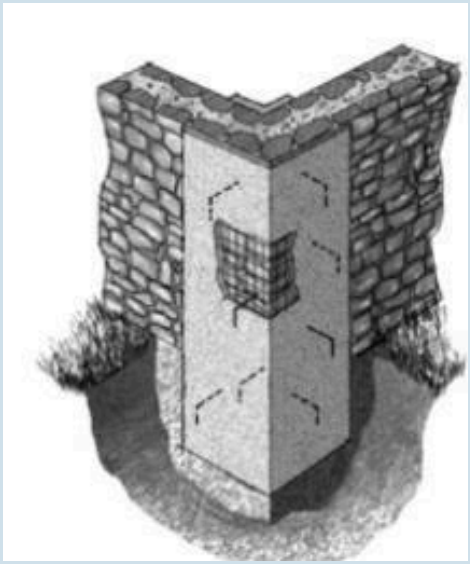
Wall to wall connection



Metal stitching



Anchored splint overlay



Through wall anchor

Through wall anchor

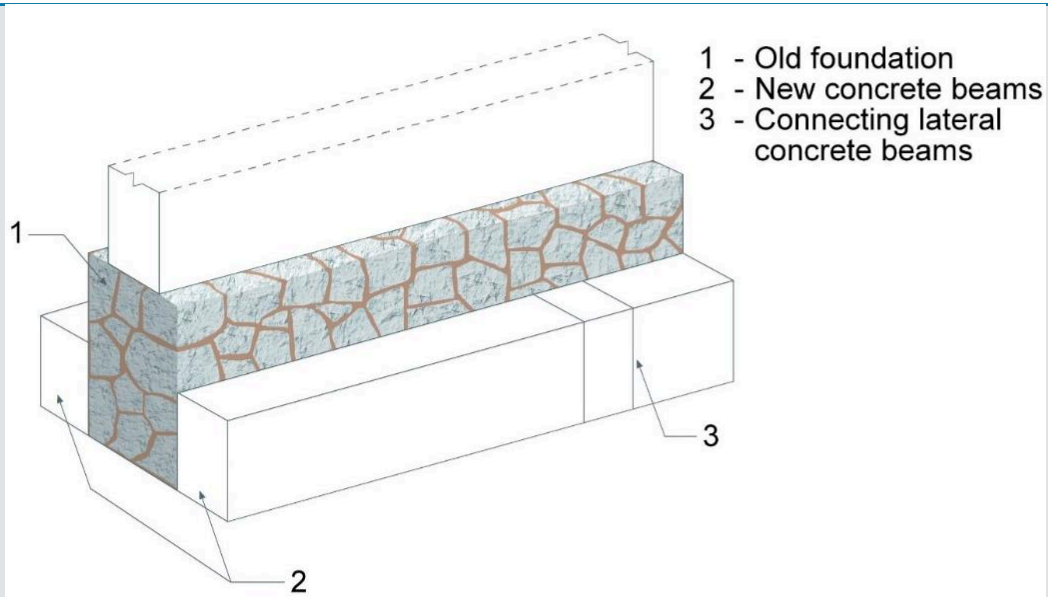
PART-B: Seismic Deficiencies and Intervention

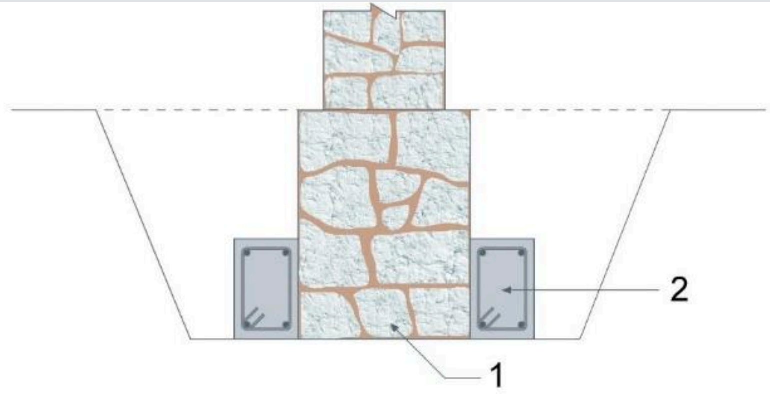
PART-B : Seismic deficiencies and intervention

This part deals with possible deficiencies in the masonry buildings and possible improvement on. Probable intervention are as follows:

1. Foundation improvement **2. Configuration and load path improvement** **3. Connection improvement between wall to wall** **4. Connection improvement between wall to floor** **5. Connection improvement between wall to roof**	**6. Stiffening of floor in their plane** **7. Stiffening of roof in their plane** **8. Tying of parapet wall** **9. Tying of gable wall** **10. Capacity improvement of structural wall with splint and bandage using:**

1. Foundation improvement

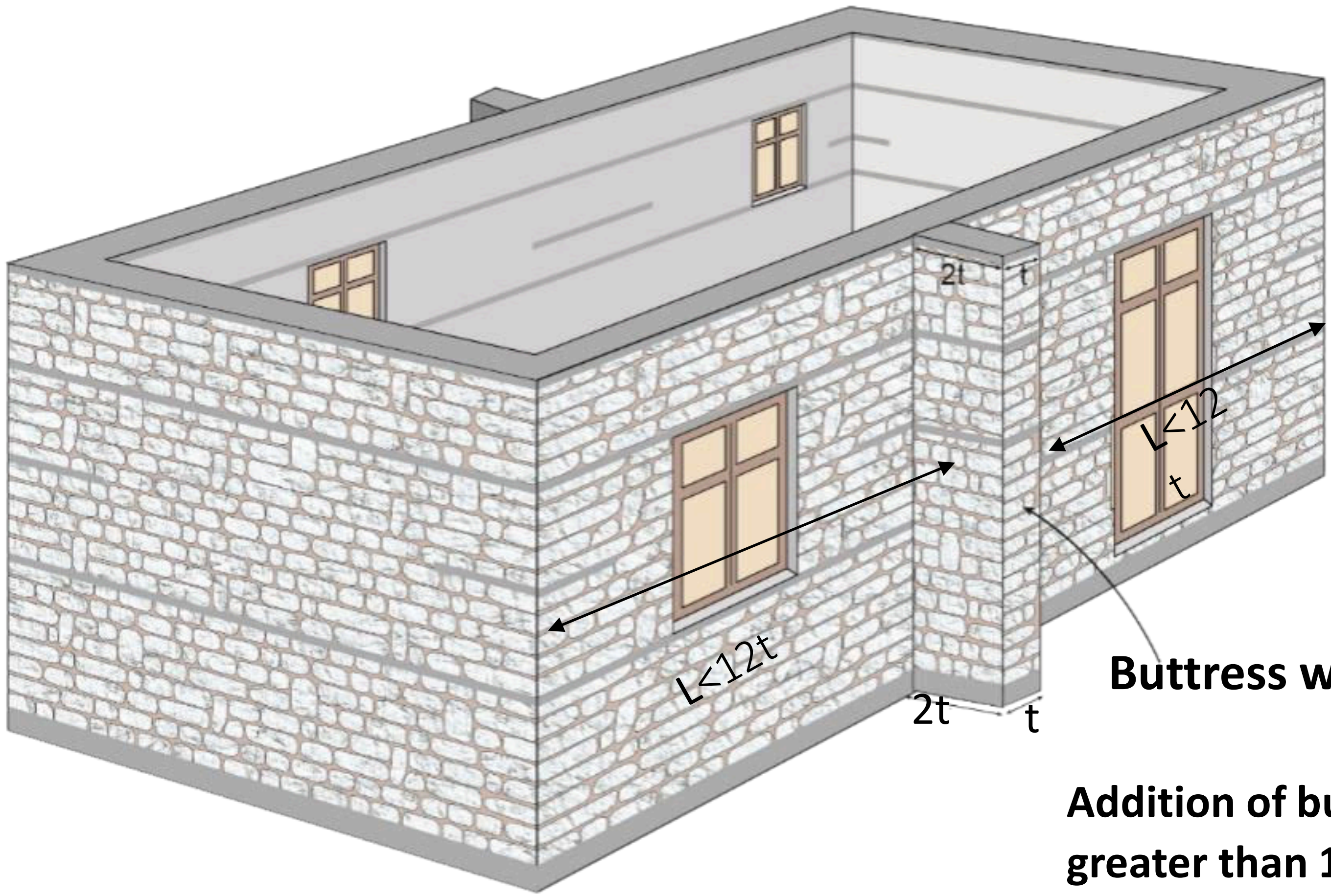




Vertical section of wall I-I

Configuration and load path improvement

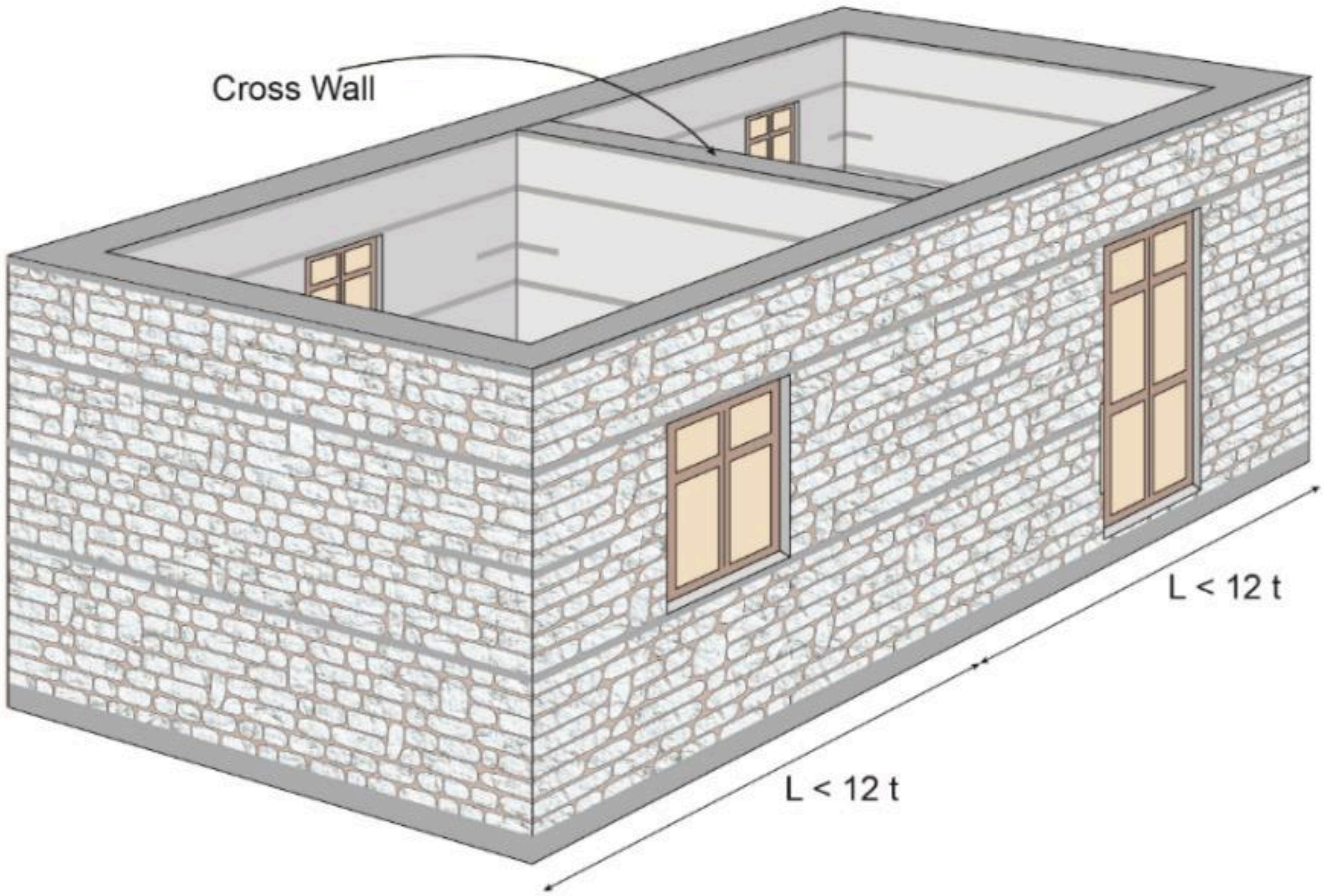
Example 1 : Construction details of Buttressing in mud mortar with flexible floor



Addition of buttress wall when length of wall is greater than 12t

Configuration and load path improvement

Example 2 : Construction details of New Cross wall in mud mortar with flexible



Addition of cross wall when length of wall is greater than 12t.

3.0 Connection improvement between wall - to – wall

Rebar anchored R C splint

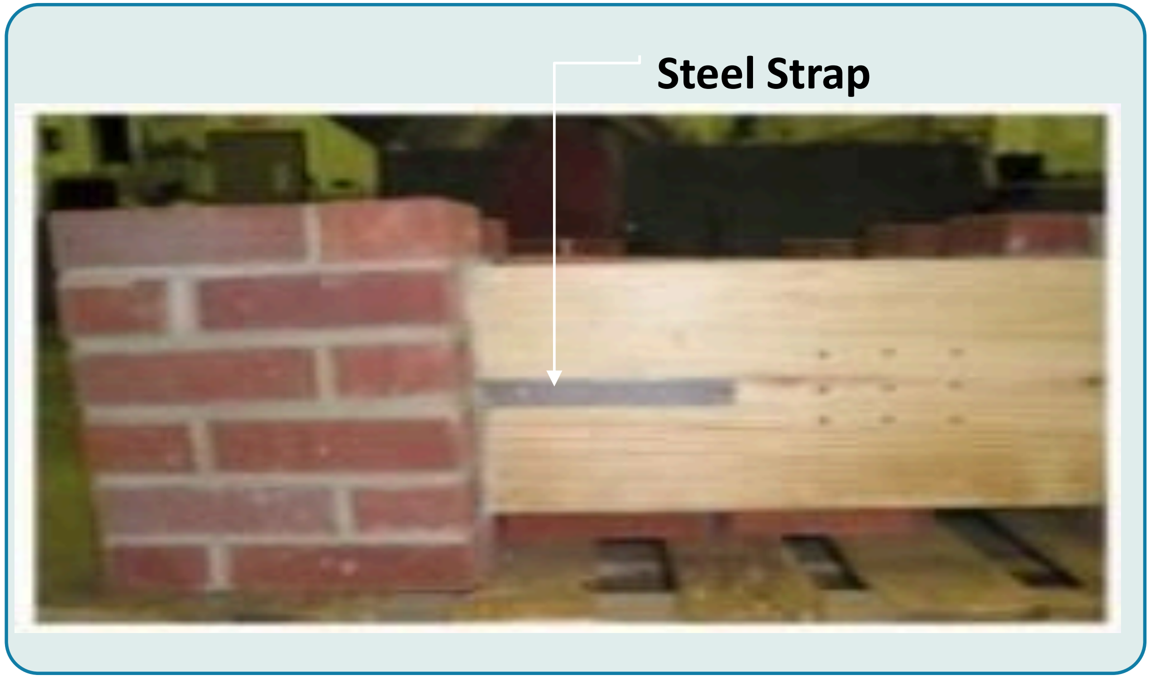
G I wire mesh anchored R C splint

3.0 Connection improvement between wall - to – wall

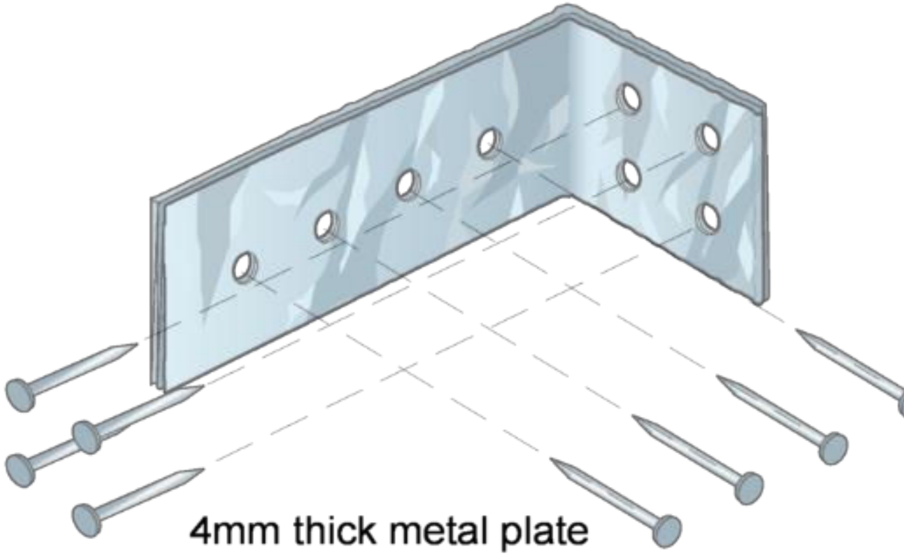
Installation of wooden vertical member from inside of the wall

4.0 Connection improvement between wall-to- floor

When it is not possible to transport concrete and steel to the site:



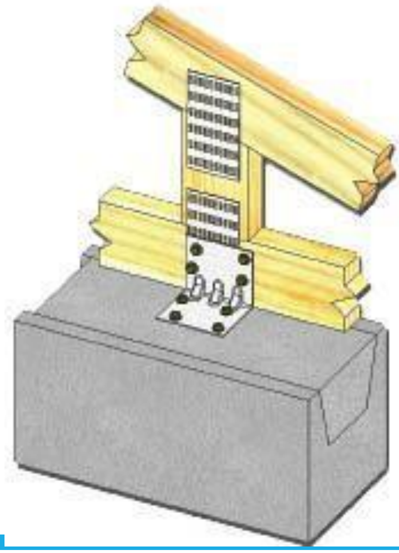
Steel Strap



Details of metal strap

5.0 Connection improvement between wall - to - roof

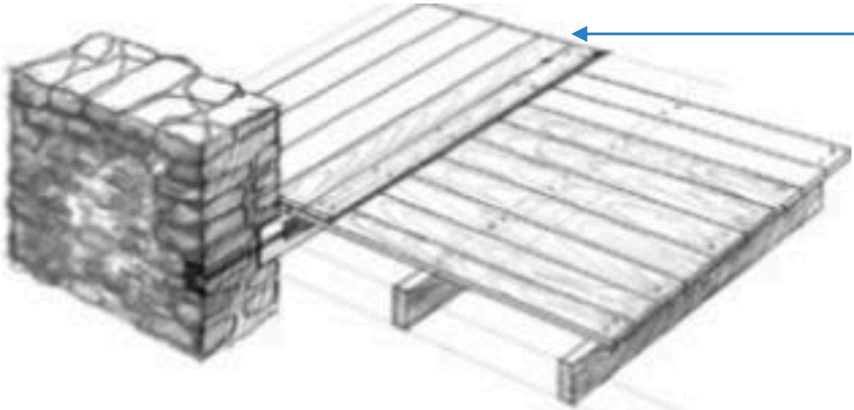
Option 1: Metal Strap at the top of wall:



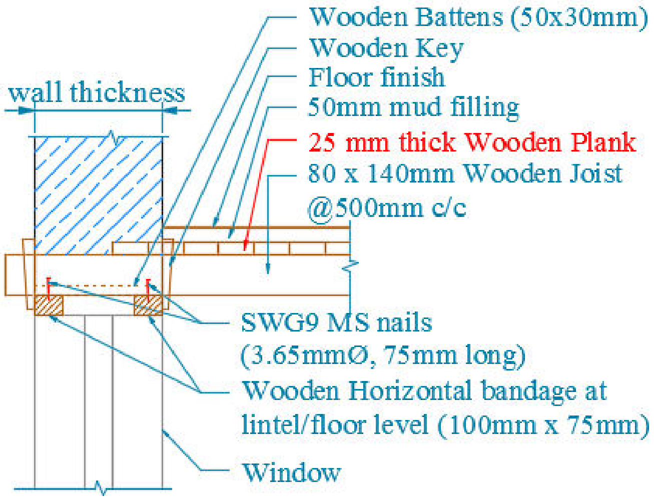
6.0 Stiffening of floors in their plane

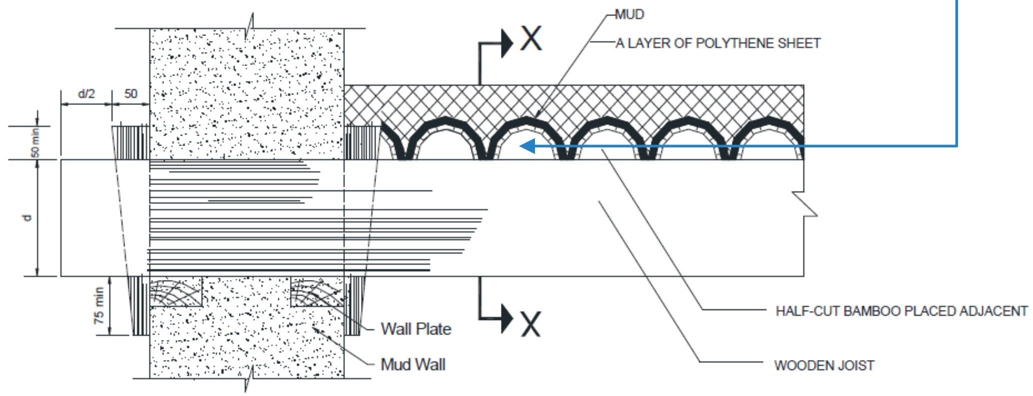
Option 1: Wooden plank overlay or half cut bamboo overlay:

25mm thick Wooden Plank overlay



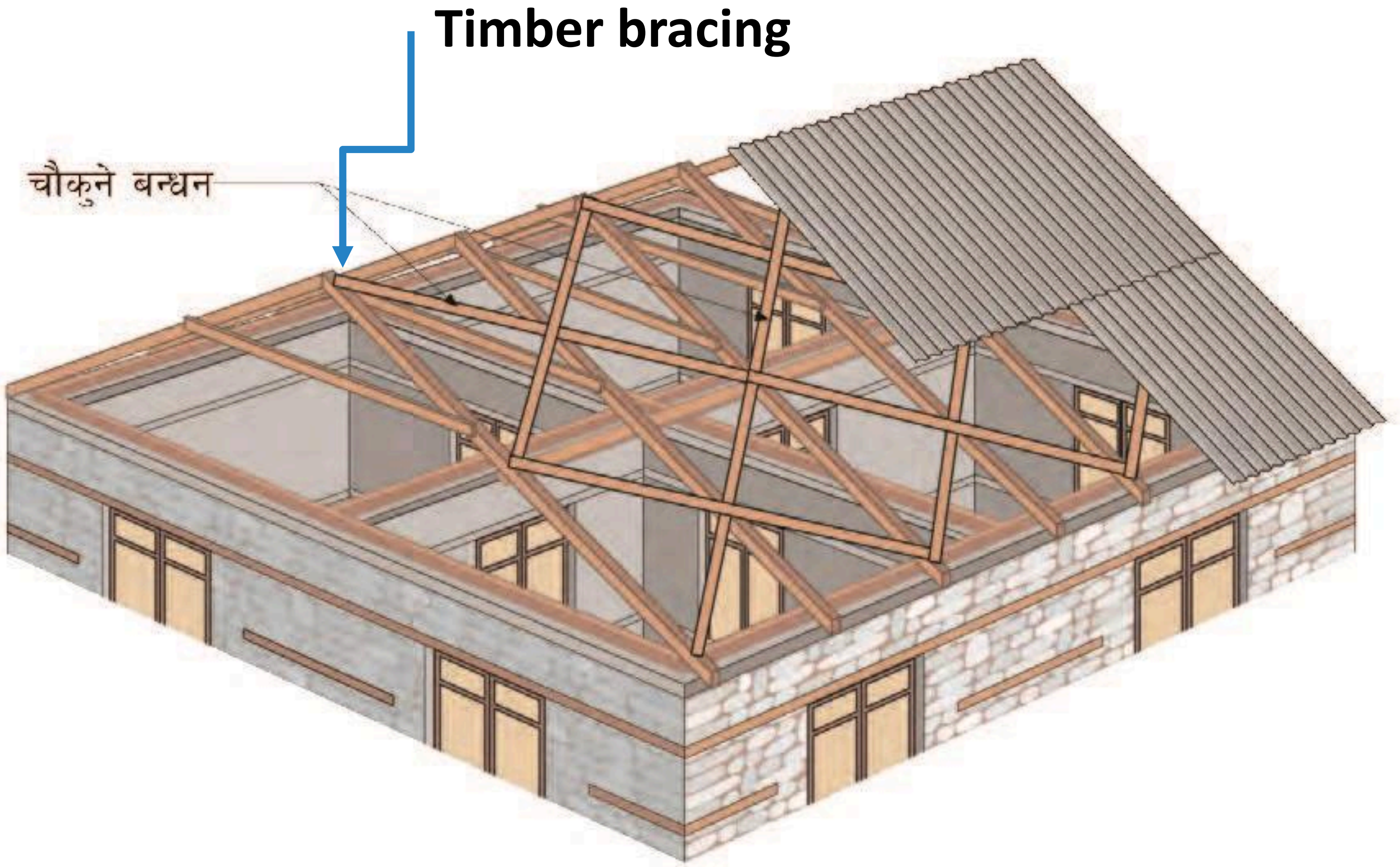
Half cut bamboo overlay





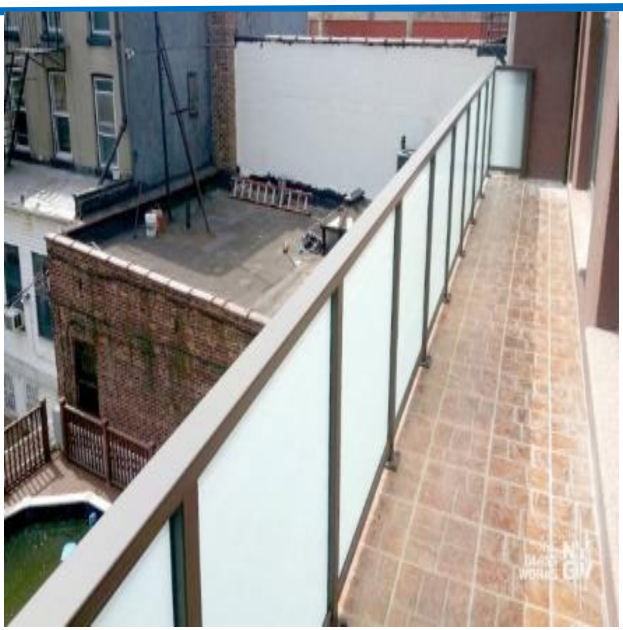
7.0 Stiffening of roofs in their plane

Option 1: Timber Bracing

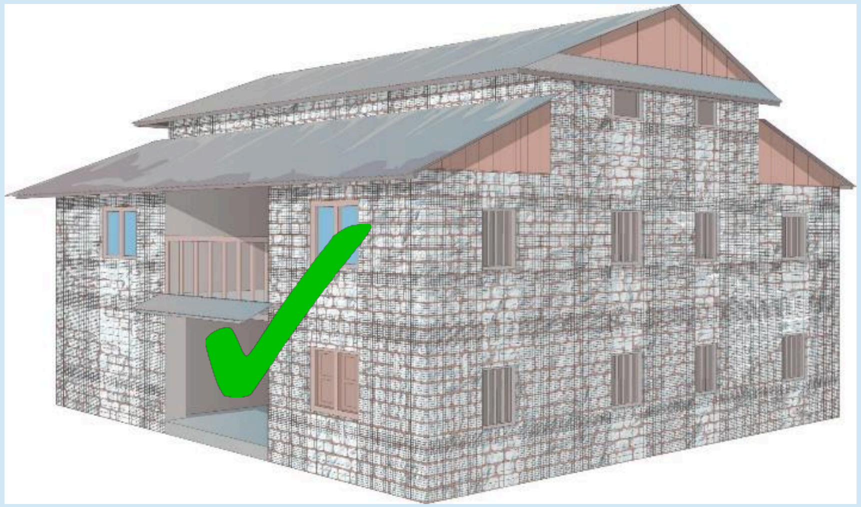


छानामा चौकुने बन्धनको विवरण

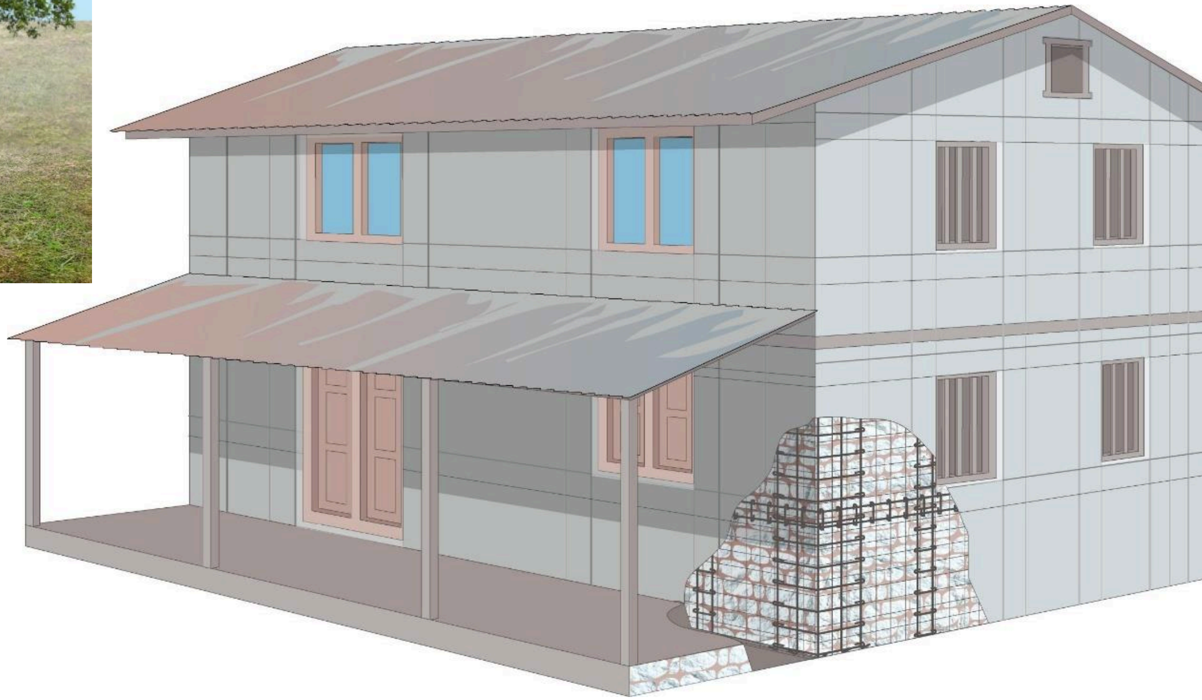
8.0 Tying of parapet walls



9.0 Tying of gable walls…

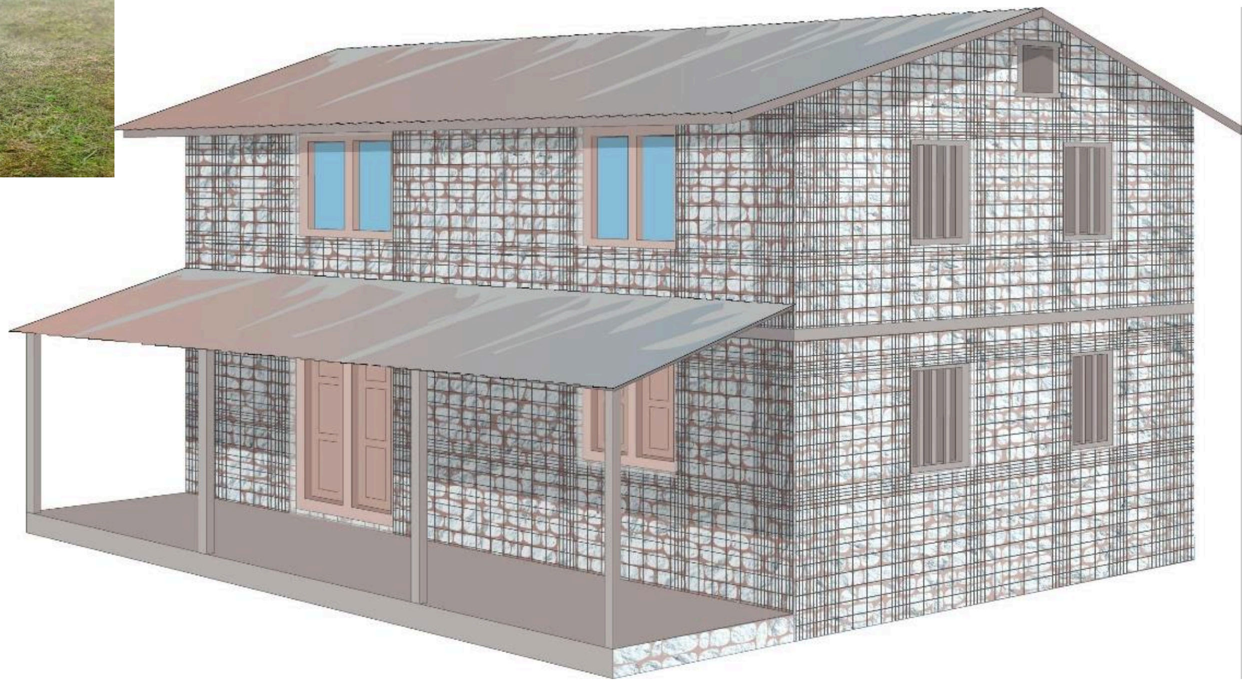


Capacity improvement of structural wall Contd.



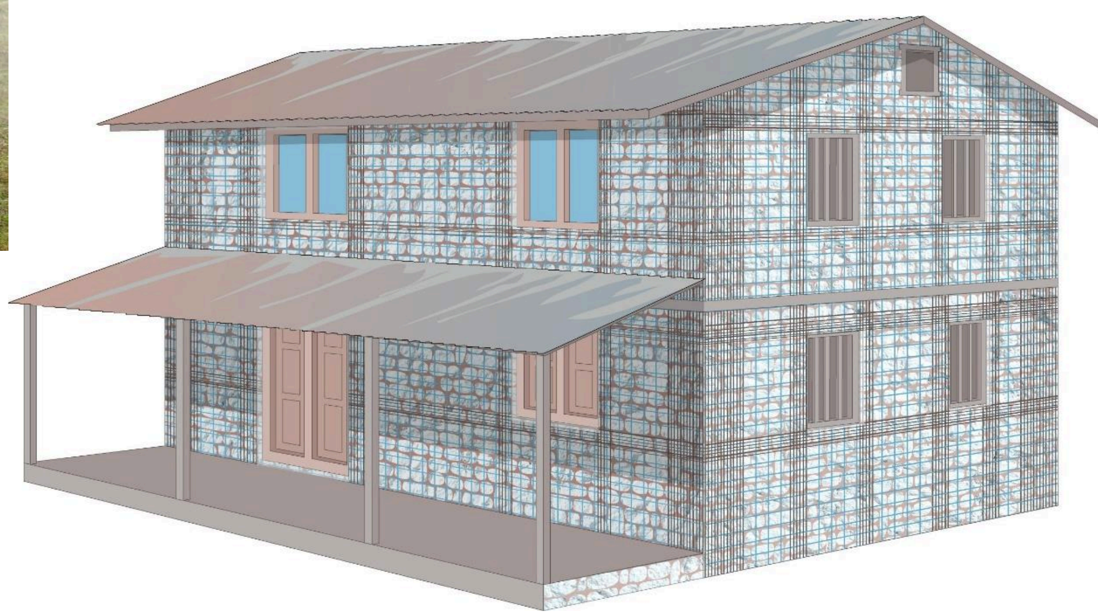
Option 1 :Retrofitting using RC splint-bandage and GI wires in remaining part

Capacity improvement of structural wall



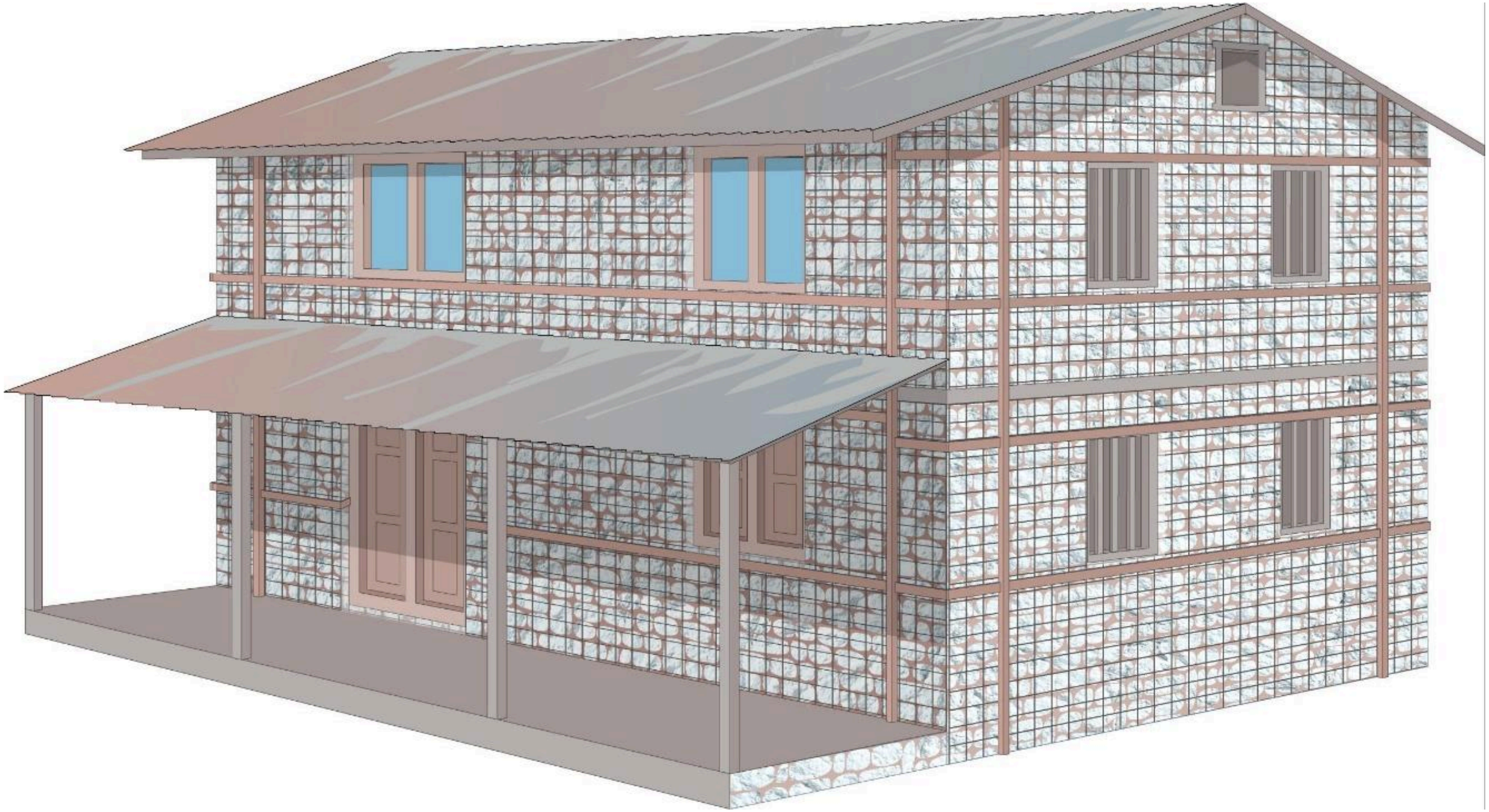
Option 2: Retrofitting using Welded wire mesh splint-bandage and GI wire mesh Jacketing

Capacity improvement of structural wall Contd.



Option 3: Retrofitting using Welded wire mesh splint and P.P. wire mesh Jacketing

Capacity improvement of structural wall Contd.



Option 4 : Retrofitting using wooden section and GI wire mesh Jacketing

PART-C : Ready to use seismic retrofit designs

PART-C : Ready to use seismic retrofit designs

This section presents summary of retrofit designs which are applicable in following cases :

- Retrofitting Stone masonry building
  1) Retrofitting stone masonry building in mud (RSMM)
  2) Retrofitting dry stone masonry building (RDSM)

- Retrofitting brick masonry building
  1) Retrofitting brick masonry building in mud with flexible floor (RBMM)
  2) Retrofitting brick masonry building in cement with flexible floor (RBMC1)
  3) Retrofitting brick masonry building in cement with rigid floor (RBMC2)

Typical description of building

- Number of storey : 2 plus attic (maximum), except RBMC2 which is three storey
- Storey height : 3.00 m (maximum)
- Total height : 7.0m (2 plus attic) and 9.0 m (three storey)
- Unsupported wall length: 5.40 m (maximum)
- Plinth area : 100.00 sq.m.
- Configuration and load path : is similar as mention in part B : Seismic deficiency and intervention)
- Redundant : Yes
- Material condition : Good or replaced with new material in case of damaged.

Option 1 : Summary of design : For RC splint & bandage

Table 3: Summary of retrofit design (applicable to RSMM, RDSM, RBMM RBMC1 : 2 plus attic storey & RBMC-2* : three storey)

S.N.	Length or Wall	Rebars Reinforcement in seismic belts with overlapping of Ld mm	
	In meter	Concrete Size (mm)	Rebar (No & diameter)
1.	≤ 5.40	300 x 40	2#10 ⌀
Note : Material grade : M20 and Fe 500 or 415 , ties 4.75 mm diameter bars @ 150 m spacing.			

Split : Rebar **in RC seismic splint with overlapping of Ld mm,**									
SN	No. of storey	Storey	Concrete size	At T-Junction		**At Corner Junction**		**At near opening**	
Concrete Grade M 20, Rebar Grade Fe 500				No	Bar ⌀ (mm)	No	**Bar ⌀ (mm)**	No	**Bar ⌀ (mm)**
1	One		200x40	3	8	3	8	2	8
2.	One plus attic	attic	200x40	3	8	3	8	2	8
		Ground	200x40	3	8	3	8	2	8
3.	Two	First	200x40	3	8(10)	3	8	2	8
		Ground	200x40	3	8 (10)	3	8	2	8
4.	Two plus attic/second	Attic/Second	200x40	3	10	3	8	2	8
		First	200x40	3	10	3	8	2	8
		Ground	200x40	3	10	3	8	2	8

Option 2 & 3 : Summary of design : For Welded G.I. Mesh splint & bandage

Table 1: Summary of retrofit design (applicable to RSMM, RDSM and RBMM)

S.N.	Length or Wall	G.I. Mesh Reinforcement in seismic belts with overlapping of 300mm		
	In meter	Gauge	No.	Wide
1.	≤ 5.40	10	12	300

Gauge, G10 = 3.25mm.

Split : G.I. Mesh Reinforcement in seismic splint with overlapping of 300mm,(applicable to RSMM, RDSM and RBMM)

SN	No. of storey	Storey	G	At T-Junction		At Corner Junction		At near opening	
				No	W	No	W	No	Wide
1	One		12	29 (10)	2X450 +T (450)	19 (10)	450 +T (450)	7 (7)	300 (300)
2.	One plus attic	attic	12	29 (10)	2X450 +T (450)	19 (10)	450 +T (450)	7 (7)	300 (300)
		Ground	12	29 (10)	2X450 +T (450)	19 (10)	450 +T (450)	7 (7)	300 (300)
3.	Two	First	14	54 (17)	2X450 +T (450)	36 (17)	450 +T (450)	9 (9)	200 (200)
		Ground	14	54 (17)	2X450 +T (450)	36 (17)	450 +T (450)	9 (9)	200 (200)
4.	Two plus attic	attic	10	56 (18)	2X450 +T (450)	36 (18)	450 +T (450)	12 (12)	300 (300)
		First	10	56 (18)	2X450 +T (450)	36 (18)	450 +T (450)	12 (12)	300 (300)
		Ground	10	56 (18)	2X450 +T (450)	36 (18)	450 +T (450)	12 (12)	300 (300)

Values in parenthesis () is for splint in side of the room and remaining value is for splint outside the room.
Gauge, G10 = 3.25mm, G12 = 2.64mm
Note : provide PP band or G.I. wires (G12) at 100 mm at centers vertically and horizontally to prevent local failures in remaining portion of the walls.

Option 4 : Summary of design : For wooden splint & bandage

Table 2: Summary of retrofit design (applicable to RSMM and RDSM)

S.N.	Length or Wall	Wooden member in seismic belts with proper overlapping.		
	In meter	size	No.	Wide
1.	≤ 5.40	38mmX75mm	2 (each face)	
Note : Connection using MS plate.				

Split : wooden member **in seismic splint with proper overlapping .**						
SN	No. of storey	Storey	Size	At T-Junction	**At Corner Junction**	**At near opening**
				No	No	No
1	One		75x75	4	3	4
2.	One plus attic	attic	75x75	4	3	4
		Ground	75x75	4	3	4
3.	Two	First	75x75	4	3	4
		Ground	75x75	4	3	4
4.	Two plus attic	attic	75x75	4	3	4
		First	75x75	4	3	4
		Ground	75x75	4	3	4
Note : Connection using MS plate. Provide G.I. wire mesh of 12 gauge @100 mm at cetres in horizontally and vertically to prevent local failures.						

Repair Process

Repair Process

a) **Action # 1:** repair (G1) minor to cracks using grouting

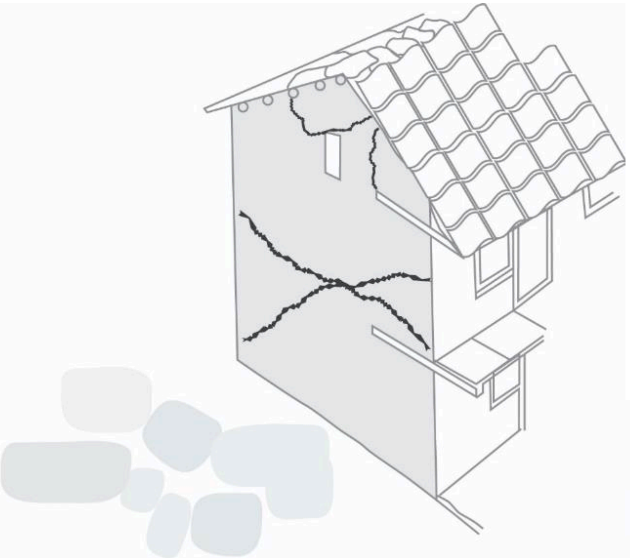
b) **Action # 2:** repair major cracks (G2 to G3) by fixing wire mesh

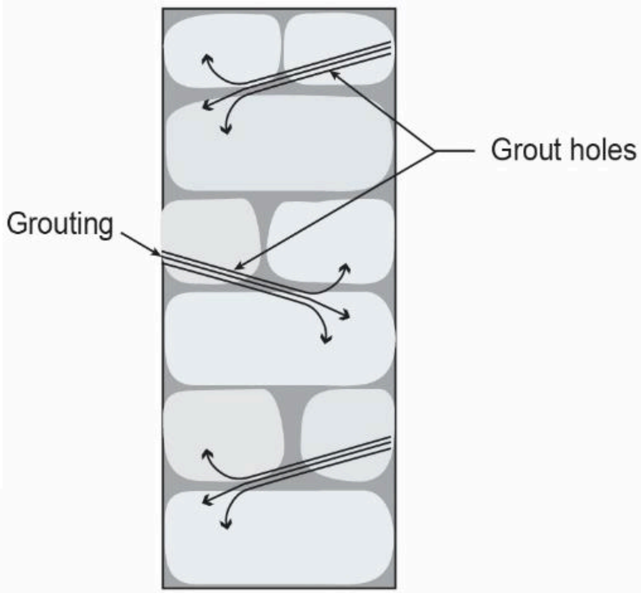
c) **Action # 3 :** repair major cracks (G2 to G3) by using stitching elements

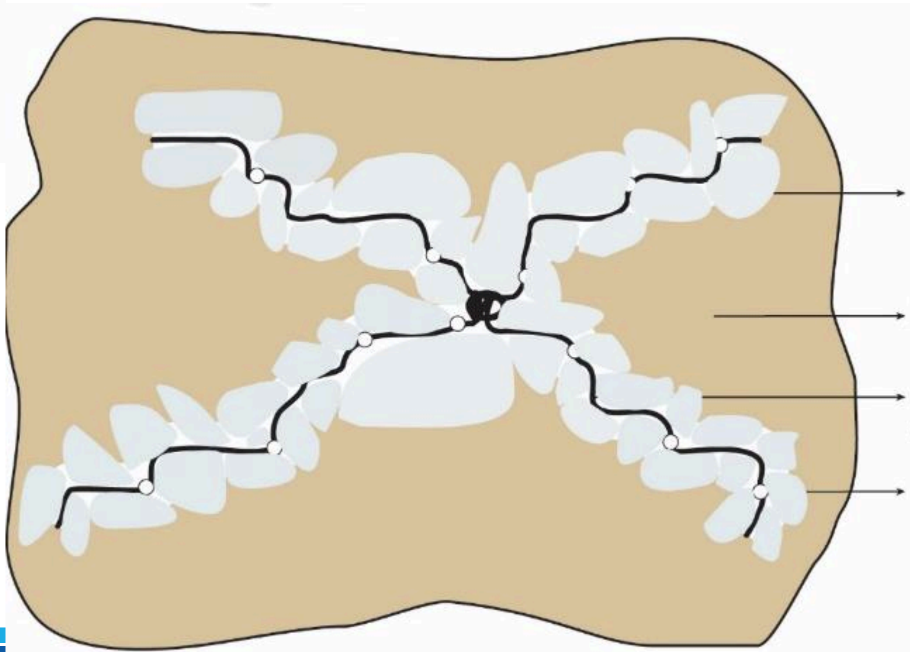
d) **Action #4**: repair of damaged wall (G2 to G3) by rebuilding

Action # 1: repair (G1) minor to cracks using grouting

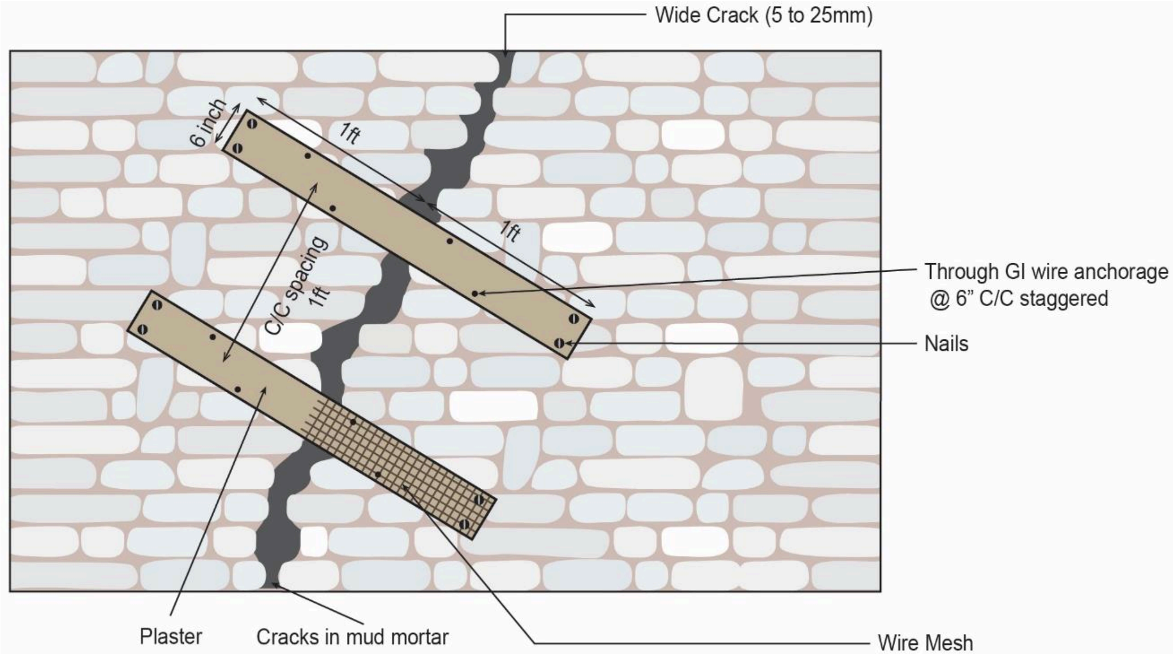
- Cracks 0.5 mm to 5mm



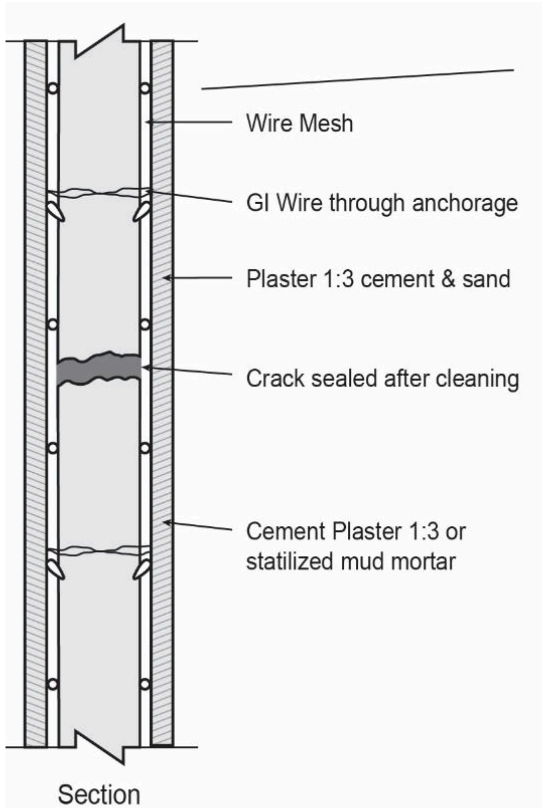




Action # 2: repair major cracks (G2 to G3) by fixing



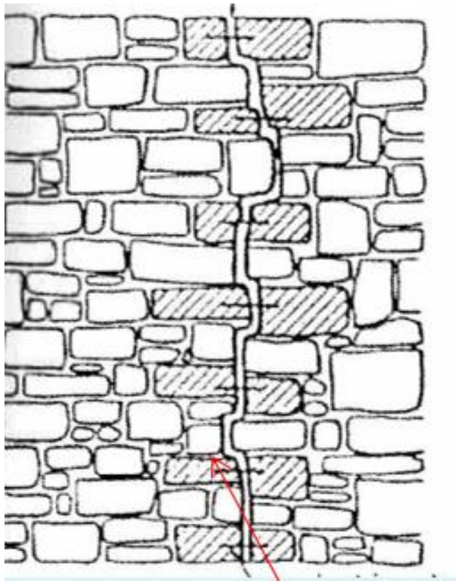
Stone Masonry : Elevation



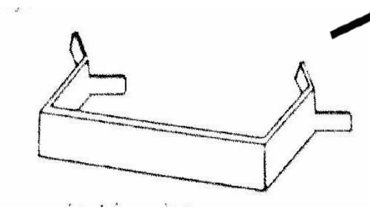
Stone Masonry : Section View

Action # 3 : repair major cracks (G2 to G3) by using stitching elements

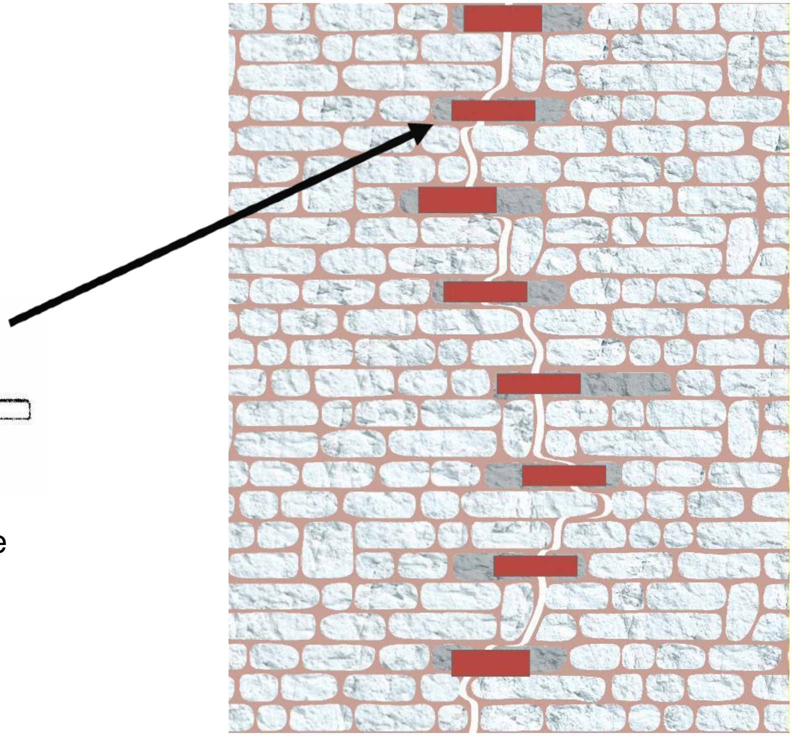
- Cracks greater than 5 mm



Removal of stones



Stitching dog (source

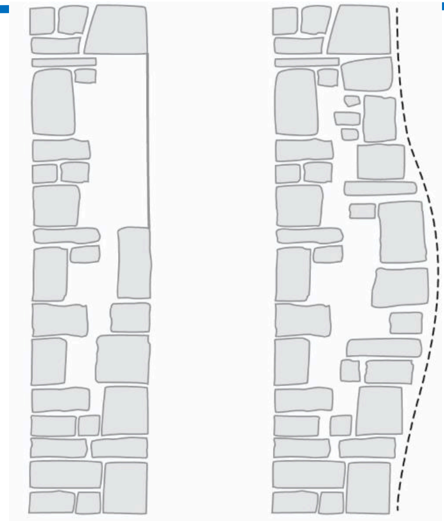


Stone Masonry

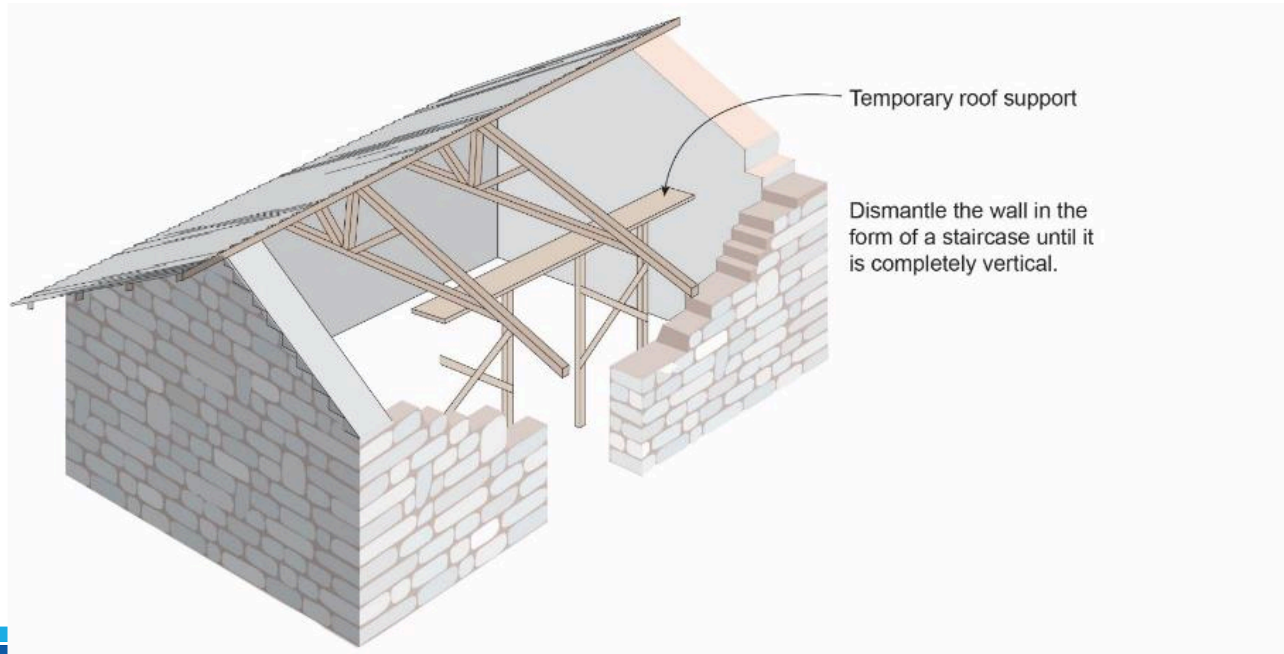
Stitching up the crack by inserting stitch on both side of wall

Action #4: repair of damaged wall (G2 to G3) by rebuilding

Stone Masonry

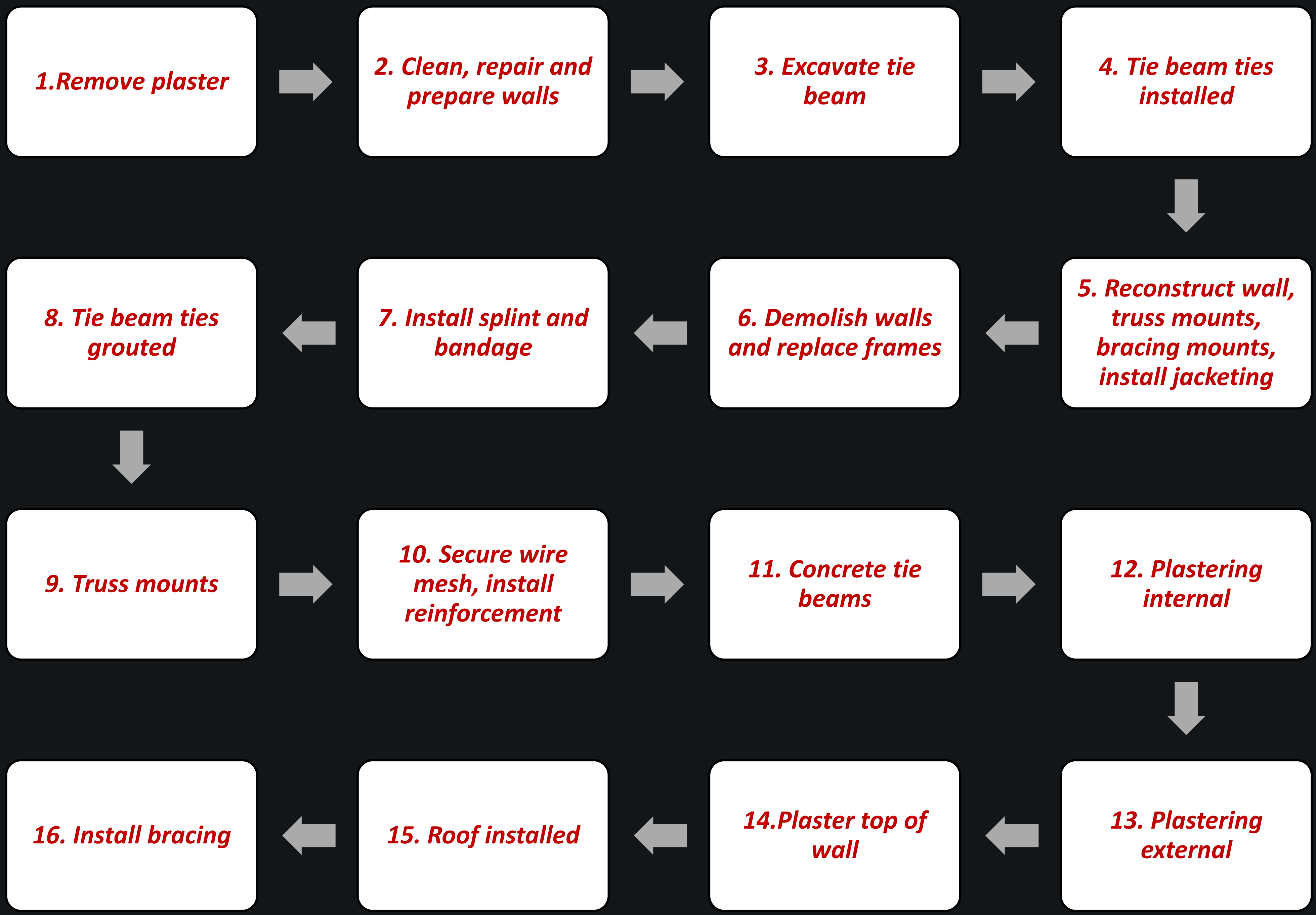


Wall that need completely removed and reconstruction



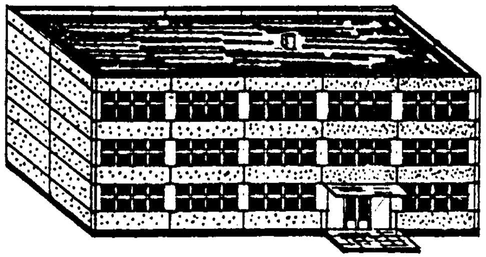
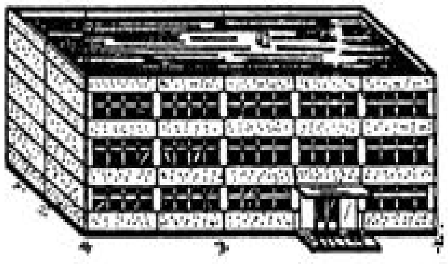
Retrofitting process

Retrofitting process

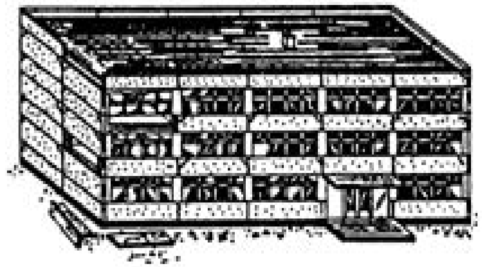
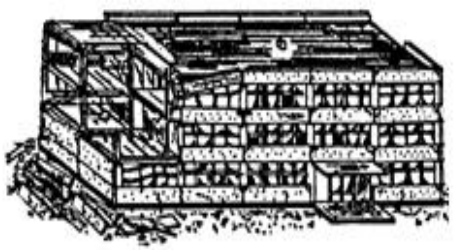


REPAIR AND RETROFITTING MANUAL For RCC STRUCTURE

Damage Grades: RC Frame Buildings

Damage Grades			Extent of Damage
DG1	Negligible-slight damage (no structural, slight non-structural)		Hair-line cracks in plaster over frame members or in walls at the base, Fine cracks in partitions and infill
DG2	Moderate damage (slight structural, moderate non-structural damage)		Cracks in columns and beams of frame and in structural walls, Cracks in partition and infill walls, fall of brittle plaster and cladding, falling mortar from joints of wall panel

Damage Grades: RC Frame Buildings

Damage Grades			Extent of Damage
DG3	Substantial to heavy Damage (moderate Structural, heavy non-structural damage)		Cracks in beam and column at the base, spalling of concrete covers, buckling of steel bars, Large cracks in partitions and infill walls, failure of individual infill panels
DG4	Very heavy damage (Heavy structural , very heavy non- structural damage)		Large cracks in structural elements with compression failure of concrete and fracture of rebars, bond failure of beam bars, tilting of columns, collapse of few columns or single upper floor

Damage Grades: RC Frame Buildings

Damage Grades			Extent of Damage
DG5	Destruction (Very heavy structural damage)		Collapse of ground floor or parts of the building

The Provision mentioned in this manual are set as only a advisory measures. For Retrofitting option of RCC Building detail Structural analysis required.

Contents

PART-A: Seismic Damage and Intervention	PART-B: Seismic Deficiencies and Intervention
1. **Material for repair and retrofitting**	**1.Major seismic Deficiencies of RCC Building**
2. Basic Repair Techniques for RCC buildings	**2. Structural Assessment Checklist-**
3. Floor Damage at Ground floor and Mitigation Work	**3. Strengthening of Footing of RCC Building**
[Key problem and repair solution]	**4. Strengthening of Column and beam**
4. RCC beam and column damage and Mitigation works	**5. Strengthening of slab/diaphragm**
[Key problem and repair solution]	**6. Strengthening of infill walls**
5 RCC Floor Slab Damage and Mitigation works	**7. Annex1:EMS Damage Grade for RCC buildings**
[Key problem and repair solution]	**8. References**
6. Other Crack Repair Technique of structural element	
5. Infill Wall damage and Mitigation works	
[Key problem and repair solution]	

1.Materials for the Repair and Retrofitting

Ordinary material	Advanced material
➢ Cement Slurry ➢ Cement Mortar ➢ Expansive Cement ➢ Quick setting Cement ➢ Gypsum Cement ➢ Steel Reinforcement ➢ GI wires ➢ Rolled Steel sections	➢ Epoxy resin/epoxy Mortar ➢ Polymer Modified Cementitious Products ➢ Fiber Reinforcement Polymers.

2. Basic Repair techniques

1.Repair of spalling or falling or minor cracks in RC members by cement slurry or mortar

2. Routing and Sealing

3. Stitching

4. Applying External Stress

5. Cement slurry Grouting/Epoxy-injection Grouting

6. Surface Overlay/ shotcrete or guniting

7. Removing buckled and yielded Reinforcement and adding new reinforcement

8. Infill wall repair

RCC Beam and Column Damage and Mitigation works

[Key Problem]

Potential damages in foundation are as follows :

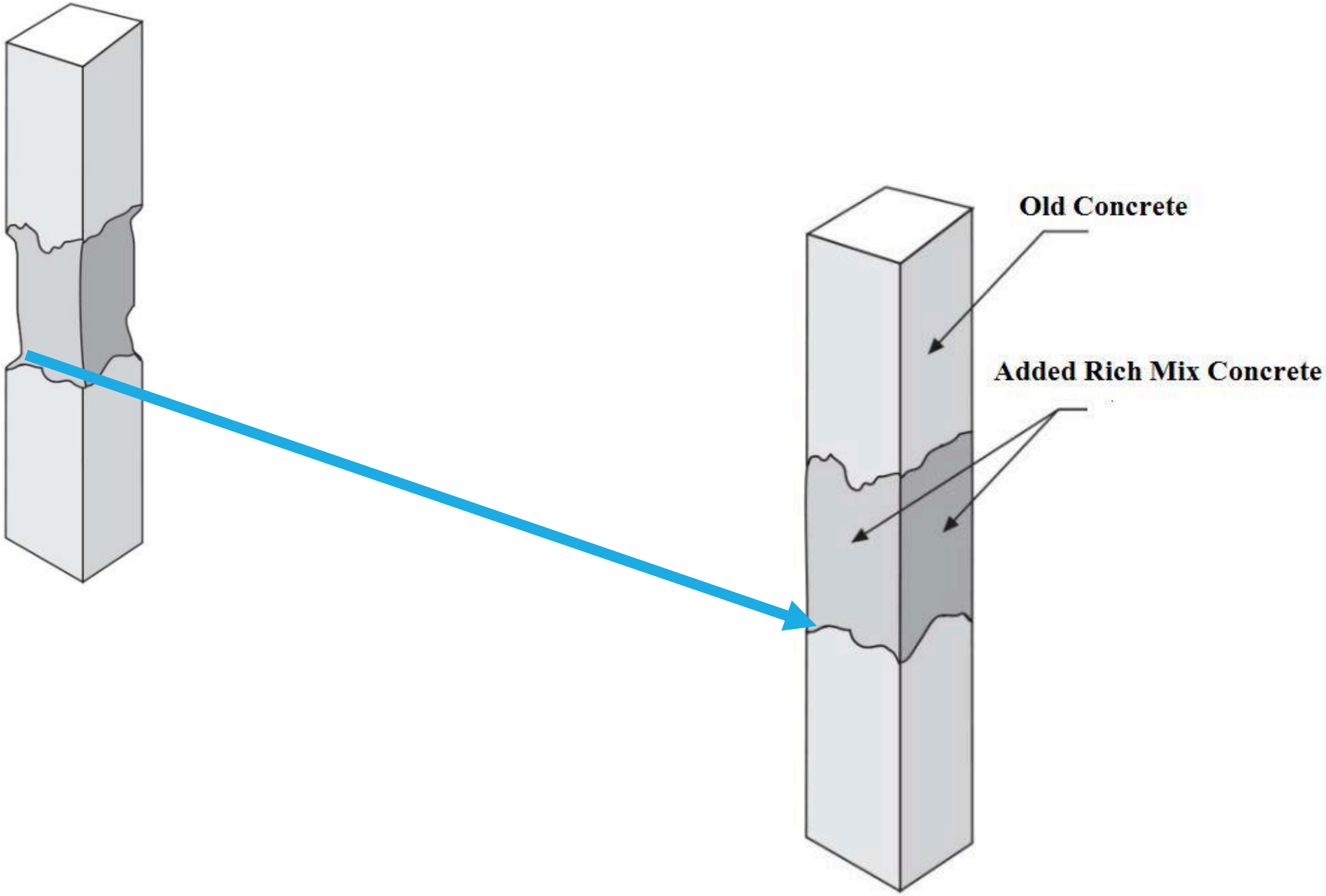
- P.1 spalling or falling of outer cover of concrete
- P.2 minor structural cracks in column and beam
- P.3 Crushing of concrete with or without buckling of bars

[Repair Solution]

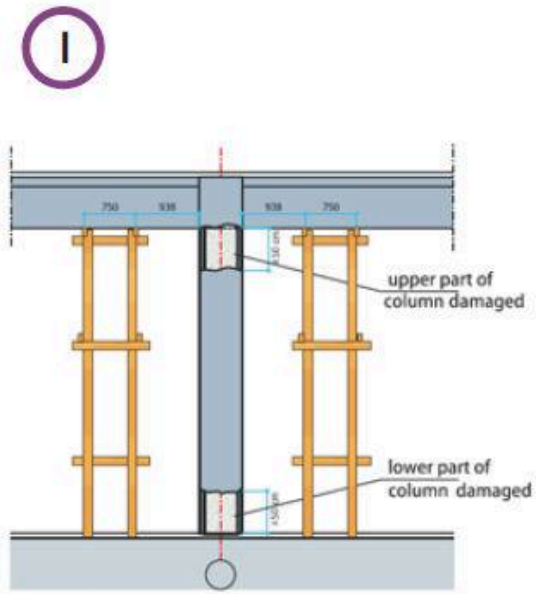
Repair solution on corresponding damages in foundation listed above, are :

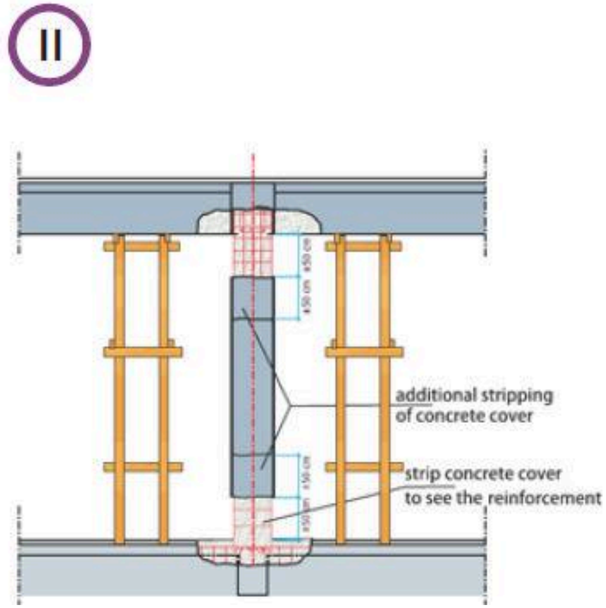
- S.1 Rich Concrete overlay:
- S.2 Epoxy Grouting in urban and semi-urban areas:
- S.3 Removing buckled and yielded reinforcement and adding new
- Reinforcement:

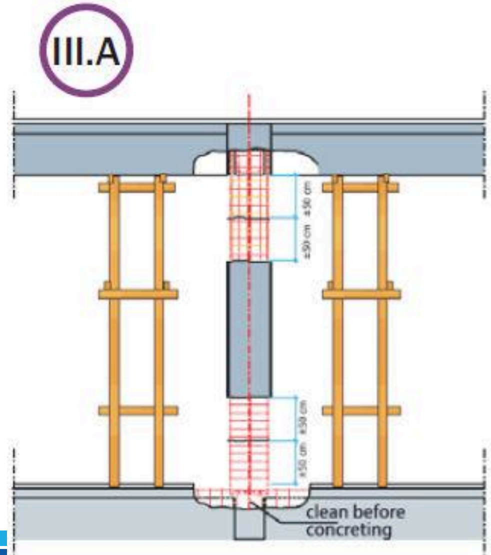
Rich mix concreting Overlay

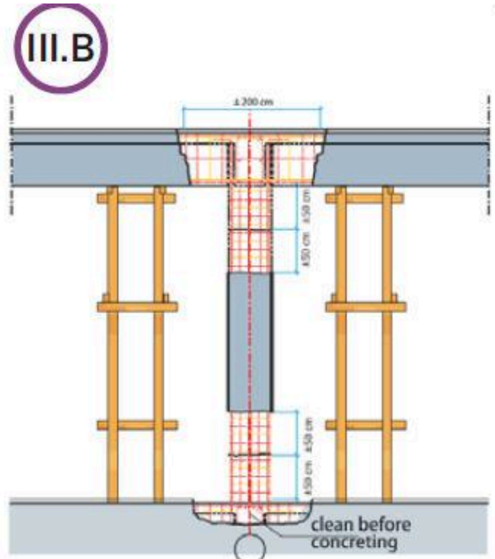


Retrofitting of top and bottom of column that are damaged

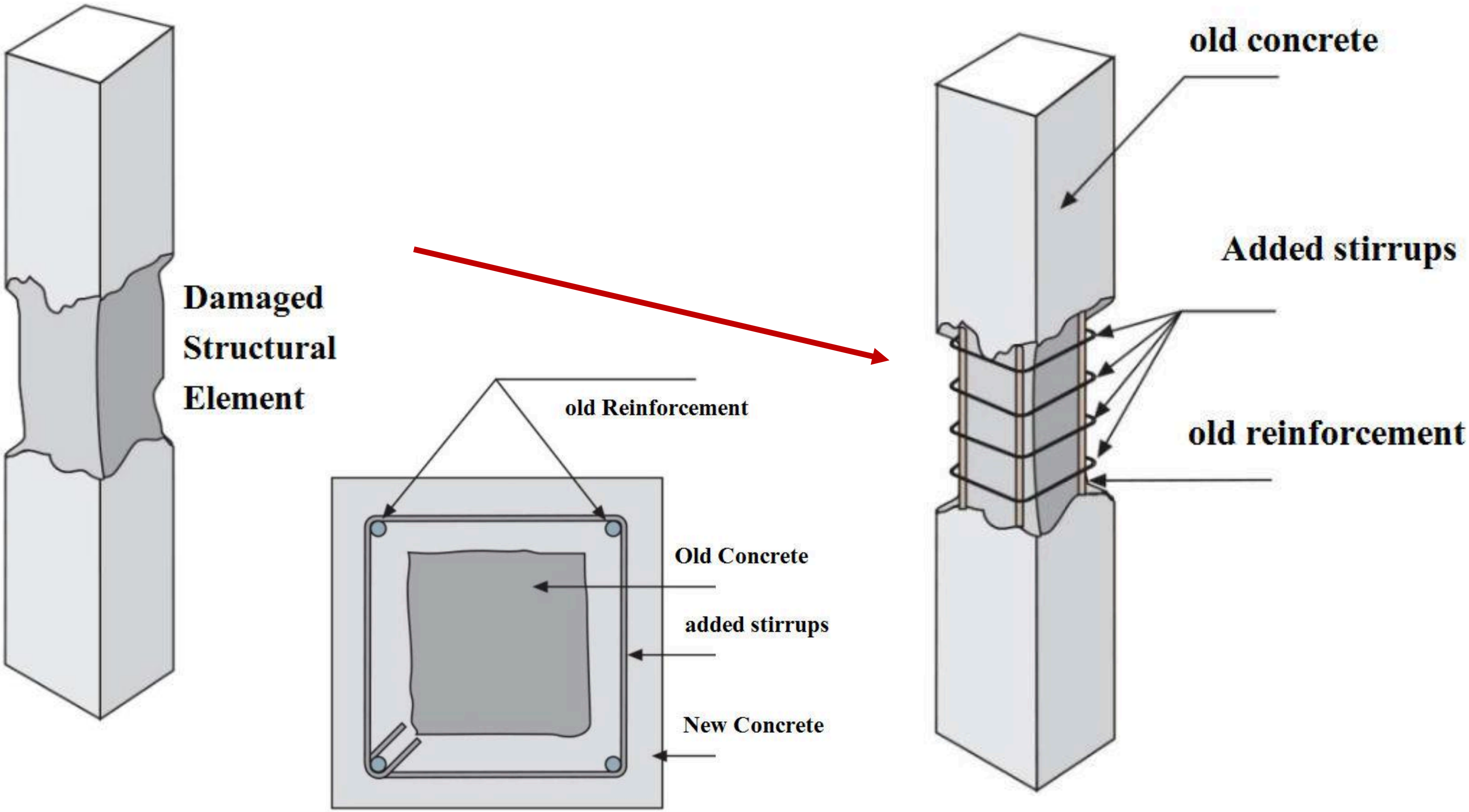








retrofitting of column by adding stirrups/reinforcement



PART-B : Seismic deficiencies and intervention

1.Major Seismic Deficiencies of RCC Building

1) Inadequate depth and size of the footing

2) Inadequate size of Structural Members

3) Inadequate amount of reinforcement on structural member

4) Missing of bands on masonry walls

5) Vulnerable Parapet walls/staircase

6) Soft/ weak Storey

Intervention for not meeting minimum Requirement

- Improvement for Foundation footing and providing
- retaining wall
- Intervention for Beam and column
- Intervention for slab
- Intervention for Masonry infill.

2. Structural Assessment Checklist

Checks	Remarks
Load Path	The frame system provides a complete load path which transfer all inertial forces in the building to the foundation. Is there clear load path?
Goemetry	Is the plan of the building is same in all stories except at basement?
Redundancy	Is There are more than two bays of frame in each direction?
weak/ soft storey	Is there are weak/soft storey?
Vertical discontinuous	Is vertical elements in the lateral force resistant element system are continuous to the foundation except for the basement floor?
Mass	Is there change in mass in adjacent floor of the building?
Torsion	Is the eccentricity of the building within the limit?
Adjacent Building	Are there adjacent building?
Short Column	Is there short column effect?
deterioration of concrete	Is there deterioration of Concrete?

3. Strengthening of Footing of RCC Building

5. Strengthening of Column and beam

1. Provide the required supporting system to the structure.

2. Remove weak concrete if exist.

3. Clean the surface and clean the rust of steel if exist.

4. Apply rust removers and rust preventers.

5. Provide additional steel **as per design** all around the section.

6. Provide required formwork.

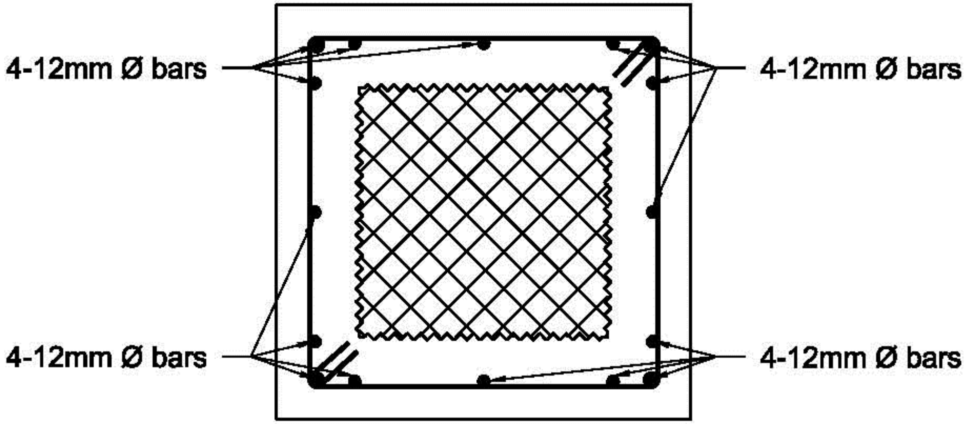
7. Provide polymer based bonding coat or rich cement bonding between old and new

concrete.

8. Place the concrete of required thickness and grade and workability admixed with

plasticizers

Concrete Jacketing

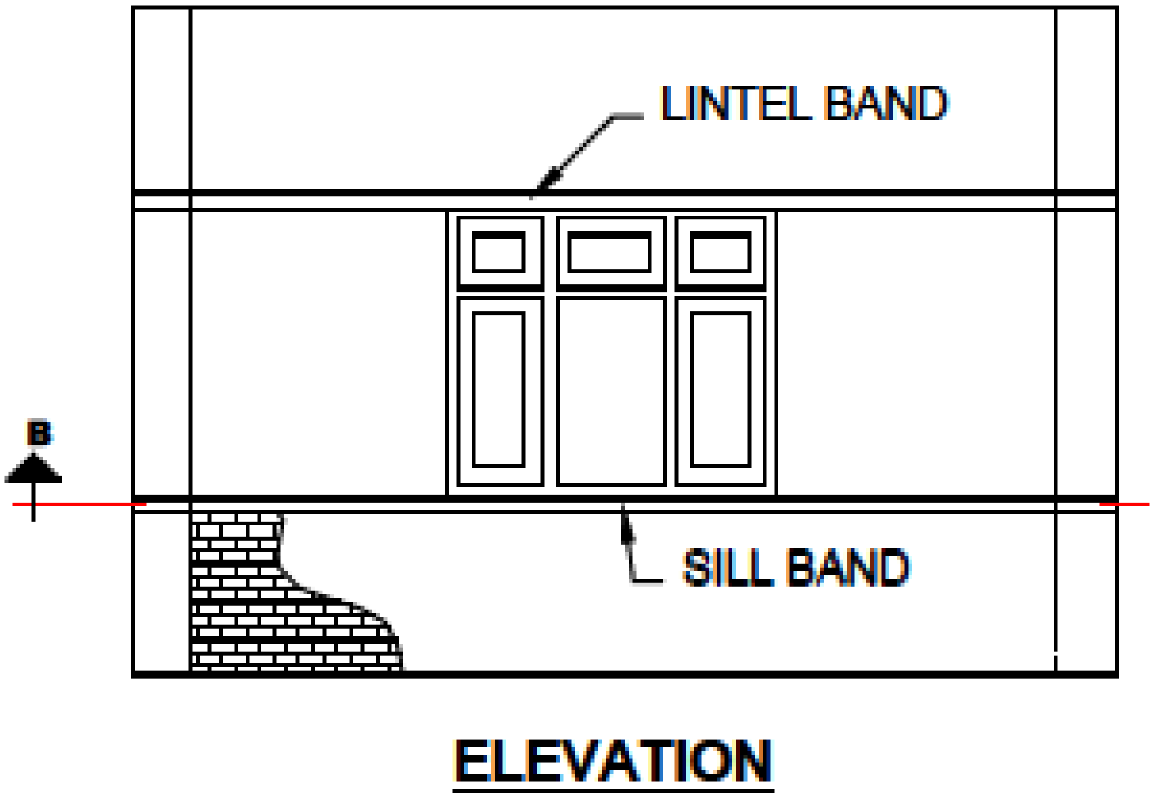


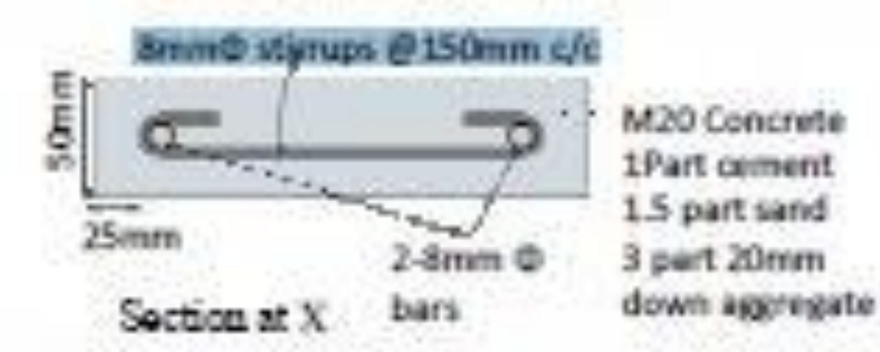
REINFORCEMENT FOR COLUMN JACKETING

Rolled Steel Jacketing

6. Strengthening of Slab/Diaphragm

7. Strengthening of infill walls





Adding sill and lintel band externally on masonry wall

अनुसूची - १७ (क) : प्रवलिकरण पूर्व प्राविधिक निरीक्षण तथा प्रमाणीकरण फाराम (प्राविधिक निरीक्षण फारम -१

नेपाल सरकार
शहरी विकास मन्त्रालय
केन्द्रीय आयोजना कार्यन्वयन इकाई

निरीक्षण फाराम (प्राविधिक निरीक्ष
प्राविधिक निरीक्षणको

घरधनी/लाभग्राहीको जानकारी		निरीक्षण मिति :	गते	महिना	वर्ष
नाम :		अनुदान सम्झौता नं.	–	– –	–
ठेगाना :	जिल्ला: गा.वि.स./न.पा.:	वडा:टोल:	जग्गाको कित्ता नं.:		
फोन / मोबाइल नं :-		अनुदान सम्झौतामा उल्लेखित बैंक खाता नं:-		बैंकको नाम :-	
खण्ड -१ :घरको जाचँको लागि दिइएको आवेदनमा भएको विवरण					
		आवेदन दर्ता नं.		आवेदन दर्ता मिति:	
	निर्माण सामग्री र प्रविधि				
	छाना र सामग्रीको निर्माण				
प्राविधिक सहायक	□छ □छैन	संस्था	□नेपाल सरकार□गैरसरकारी संस्था( )		
तालिम प्राप्त डकर्मी प्रयोग गरिएको पहिला पुनर्निर्माणमा परेको र पछि प्रवलीकरण गर्न चाहेमा			□छ	□छैन	
खण्ड -२ : विस्तृत damage and technical suggestion प्राविधिक विवरण					

क)	जगमा क्षति		छ		छैन
	क्षतिको स्तर		सामान्य		धेरै
सुझावहरु	....				
ख)	सँरचनागत अवयवहरुमा क्षति		छ		छैन
	क्षतिको स्तर		सामान्य		धेरै
सुझावहरु (पिलर वाला घर)	पिलर .... बिम .... बिम र पिलरको जोनि... तल्ला .... छाना				
गाह्रो वाला घर	गारो .... बिम .... तल्ला .... छाना .... संयोजकता (connectivity)..				

अनुसूची १७ (ख)-प्रवलिकरण पश्चात् प्राविधिक निरीक्षण तथा प्रमाणीकरण फाराम (प्राविधिक निरीक्षण-२)

नेपाल सरकार
शहरी विकास मन्त्रालय
केन्द्रीय आयोजना कार्यन्वयन इकाई

निरीक्षण फाराम (प्राविधिक निरीक्षण-२)
प्राविधिक निरीक्षणको लागि

घरधनी/लाभग्राहीको जानकारी		निरीक्षण मिति :	गते	महिना	वर्ष
नाम :		अनुदान सम्झौता नं.	–	– –	–
ठेगाना :	जिल्ला: गा.वि.स./न.पा.:	वडा:टोल:	जग्गाको कित्ता नं.:		
फोन / मोबाइल नं :-		अनुदान सम्झौतामा उल्लेखित बैंक खाता नं:-		बैंकको नाम :-	
खण्ड -१ :घरको जाचँको लागि दिइएको आवेदनमा भएको विवरण					
स्वीकृत नक्सा-डिजाइन मध्येको भए		आवेदन दर्ता नं.		आवेदन दर्ता मिति:	
	निर्माण सामग्री र प्रविधि				
	छाना र सामग्रीको निर्माण				
प्राविधिक सहायक	□छ □छैन	संस्था	□नेपाल सरकार□गैरसरकारी संस्था( )		
तालिम प्राप्त डकर्मी प्रयोग गरिएको			□छ	□छैन	
खण्ड -२ : विस्तृत प्राविधिक विवरण					

विवरण		सुझावहरु	सुझावहरु पालना भएको			
क) जग		१.. २.. ३..		छ		छैन
ख) सँरचनागत अवयवहरु						
	पिलर	१.. २.. ३..				
	बिम	१.. २.. ३..				
	बिम र पिलरको जोनि	१.. २.. ३..				
	तल्ला	१.. २.. ३..				

Thank you